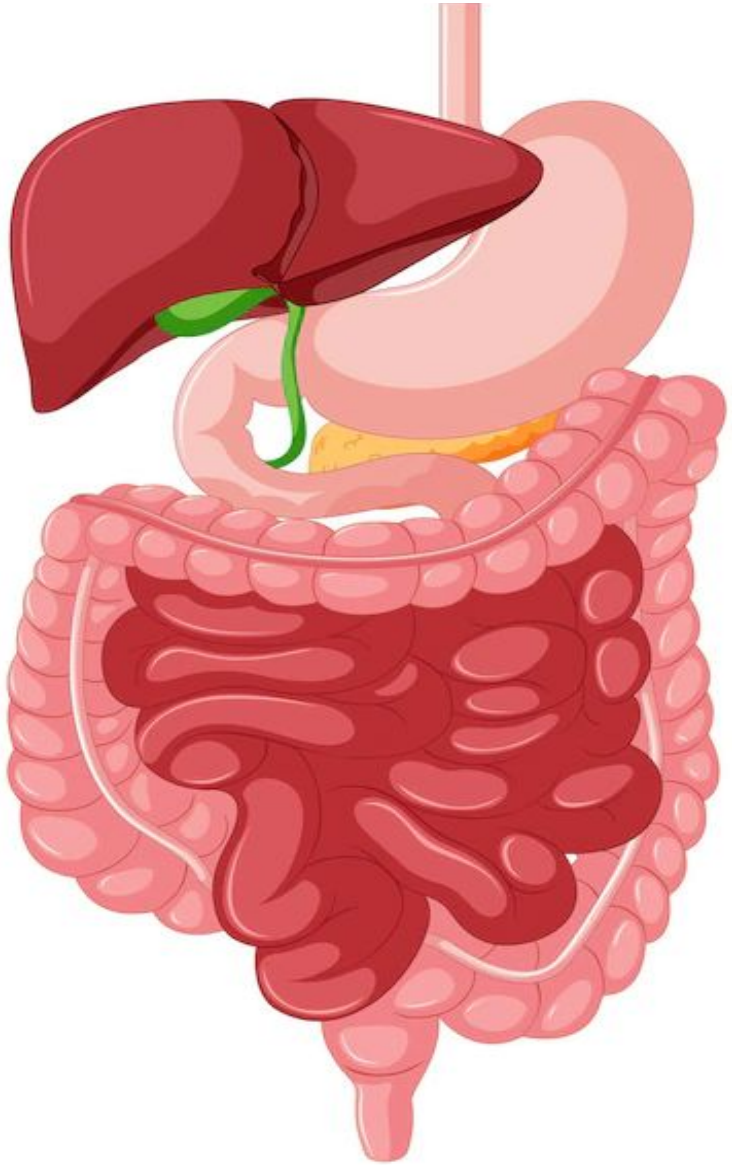
Overview of GIT (Gastro-intestinal Tract)

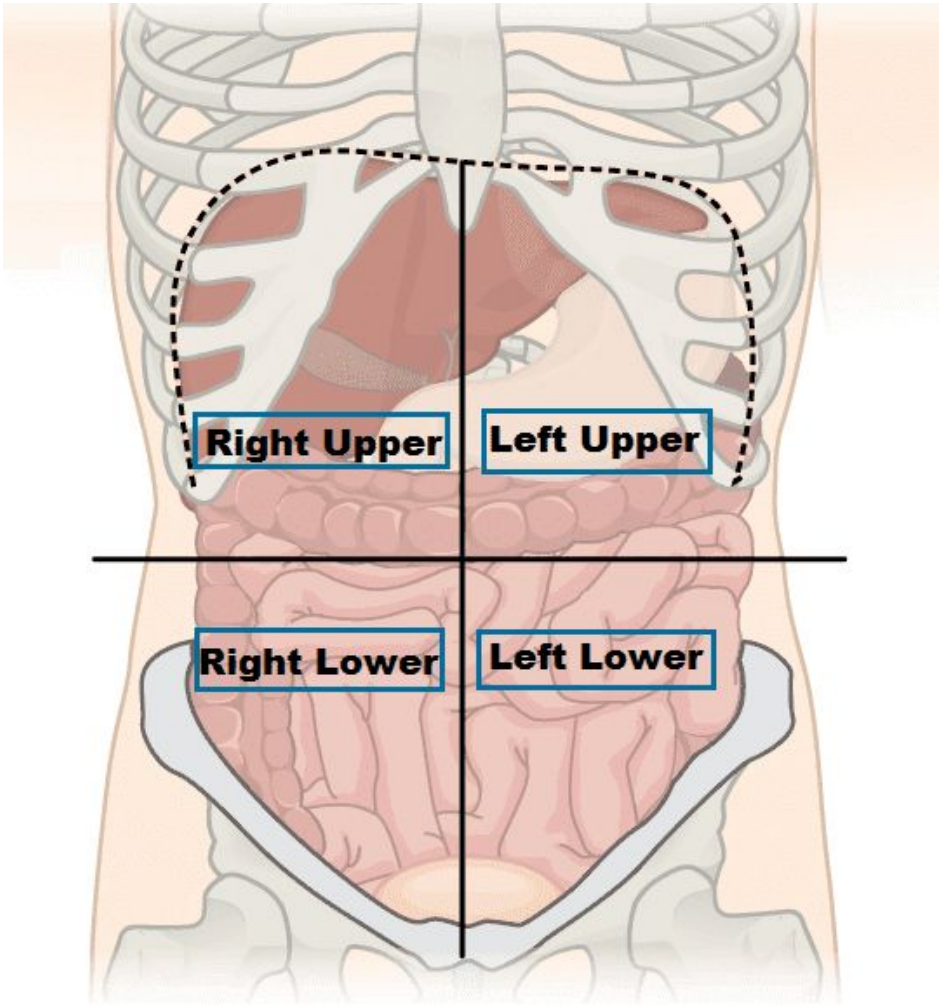


Topographical Divisions of Abdomen:

- The topographical divisions of abdomen are used to describe the location of abdominal organs and the pain associated with abdominal problems.
- The two patterns are often used:

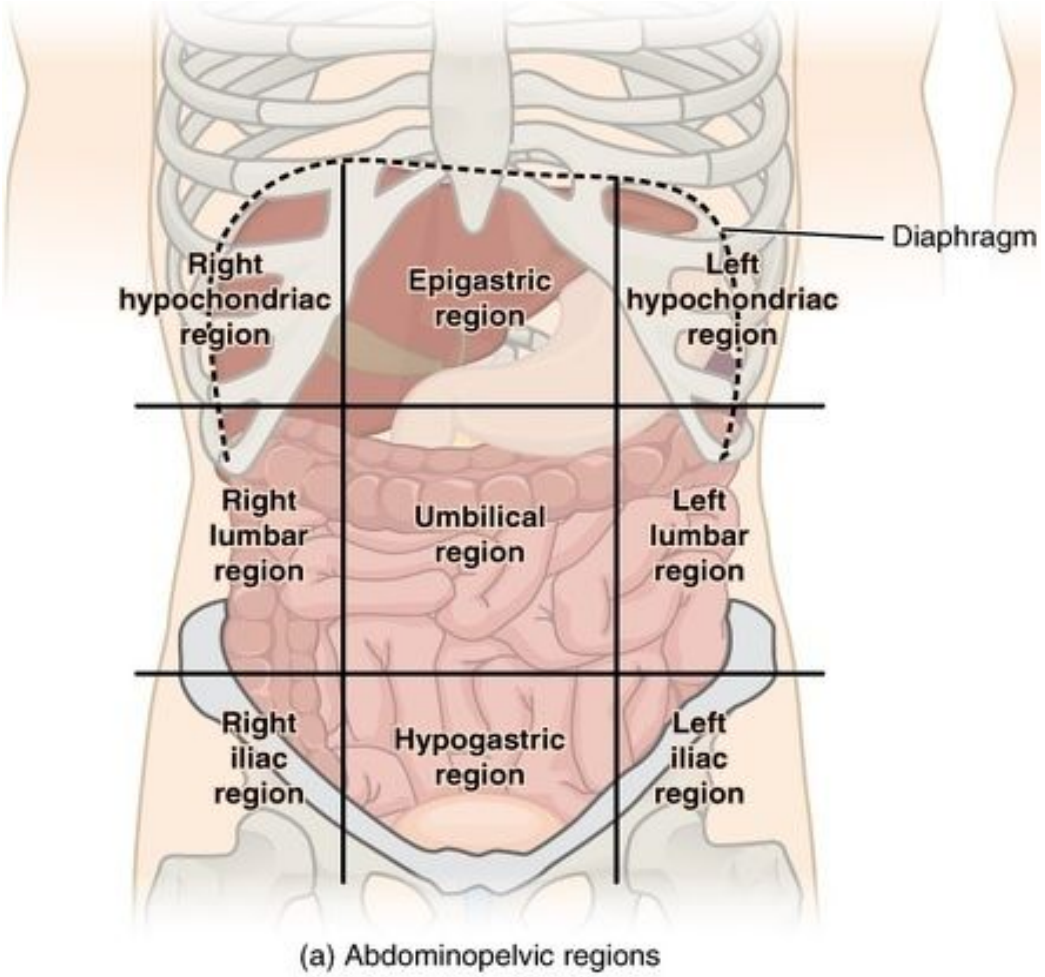
1) Four- quadrant pattern:

- A horizontal ***transumbilical plane*** passes through the umbilicus and the intervertebral disc between L3 & L4 vertebrae.
- The transumbilical plane intersects with the ***vertical median plane*** dividing the abdomen into four quadrants: ***the upper right, the upper left, the lower right***, and ***the lower left.***



2) Nine-region pattern: This pattern is based on two vertical and two horizontal lines.

- The **vertical planes** pass from the midpoint of clavicle to a point midway between anterior superior iliac spine and pubic symphysis.
- The ***transpyloric plane*** passes through the tip of 9th costal cartilages on both sides and it lies at the level of L1 vertebra
- The ***subcostal plane*** lies immediately inferior to the lower border of 10th costal cartilage, and corresponds to the level of body of L3 vertebra.
- The ***intertubercular plane*** connects the tubercles of iliac crest and passes through the upper border of L5 vertebra.



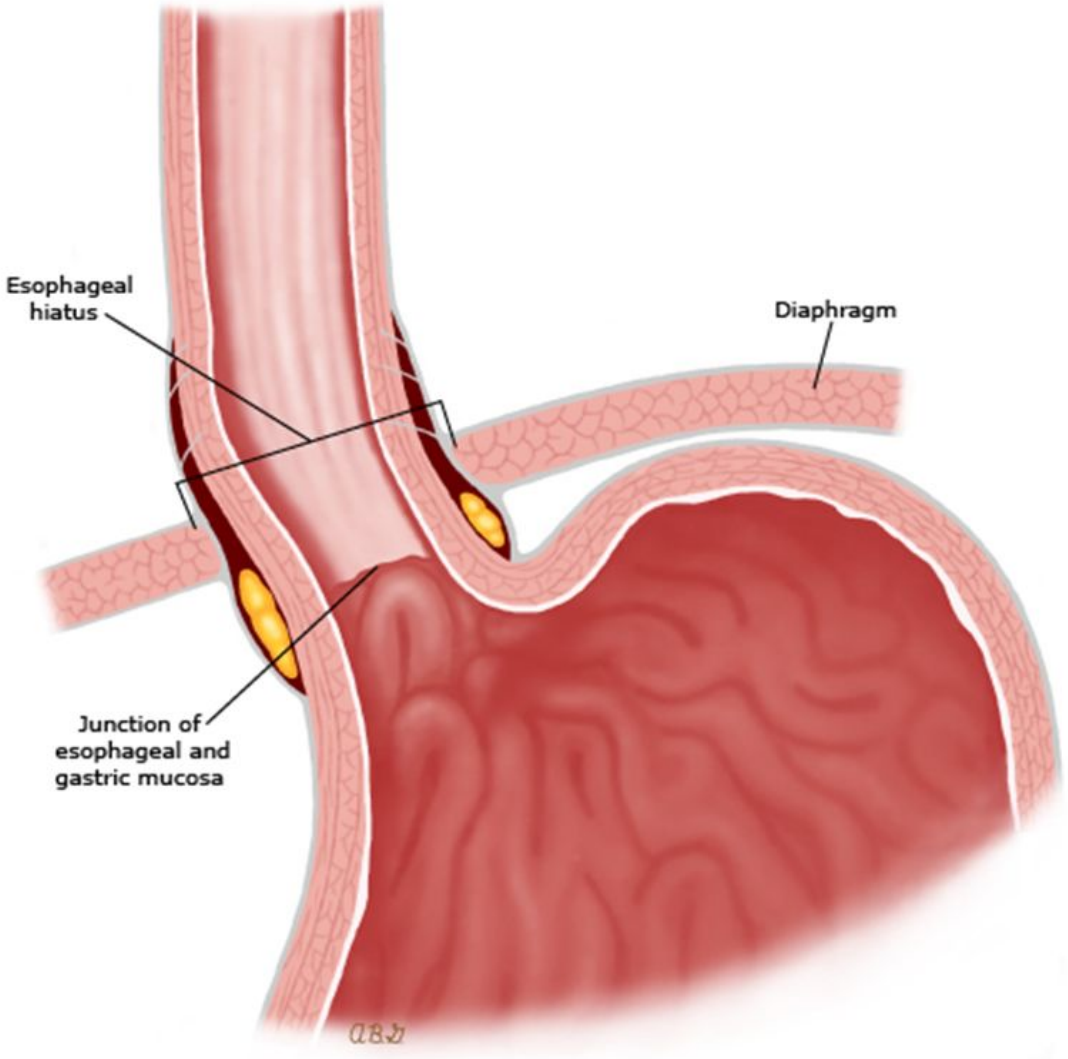
(a) Abdominopelvic regions

Introduction to GIT:

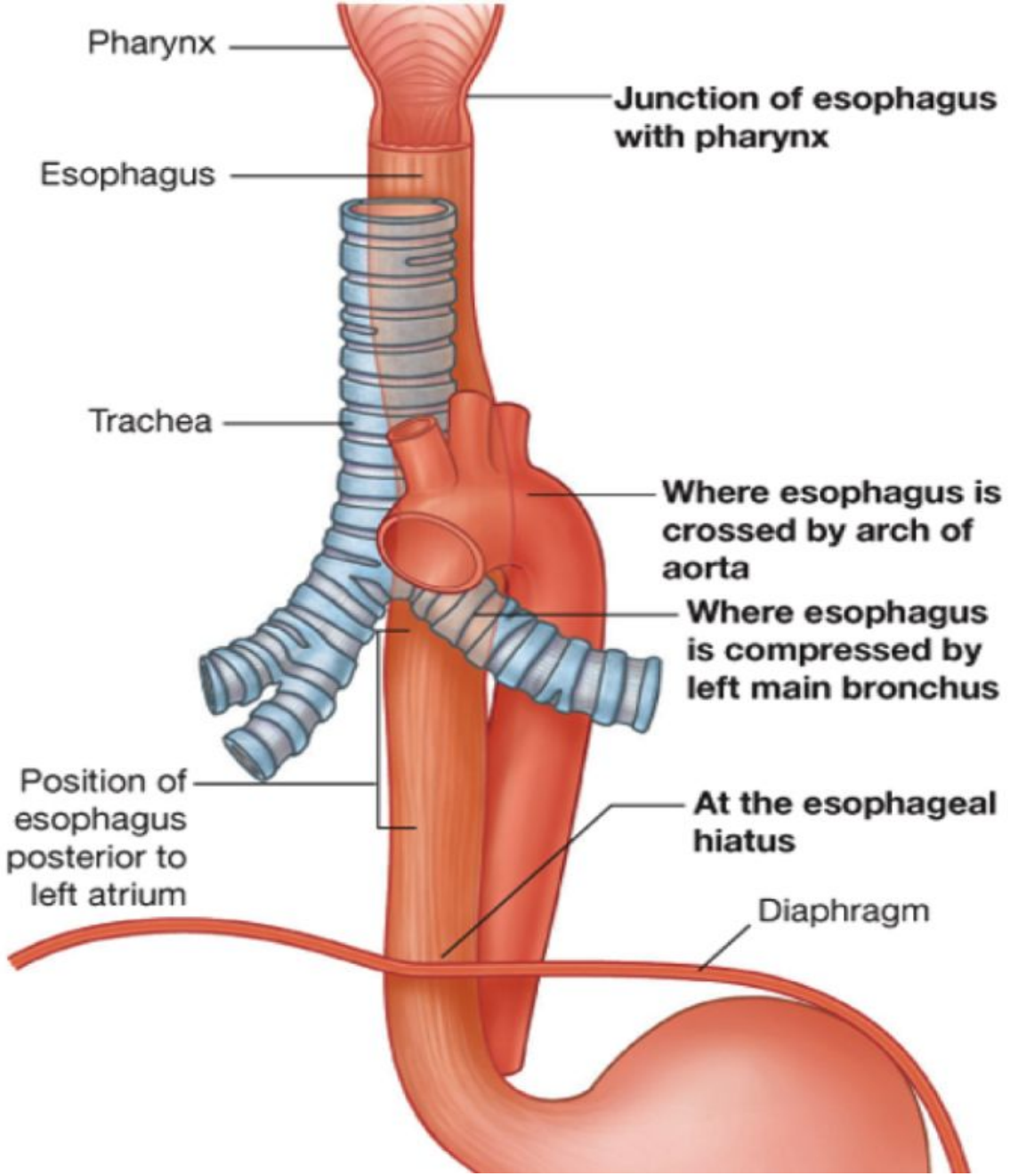
- GI tract includes the stomach, small intestine, large intestine, and other accessory organs.

Abdominal part of Esophagus:

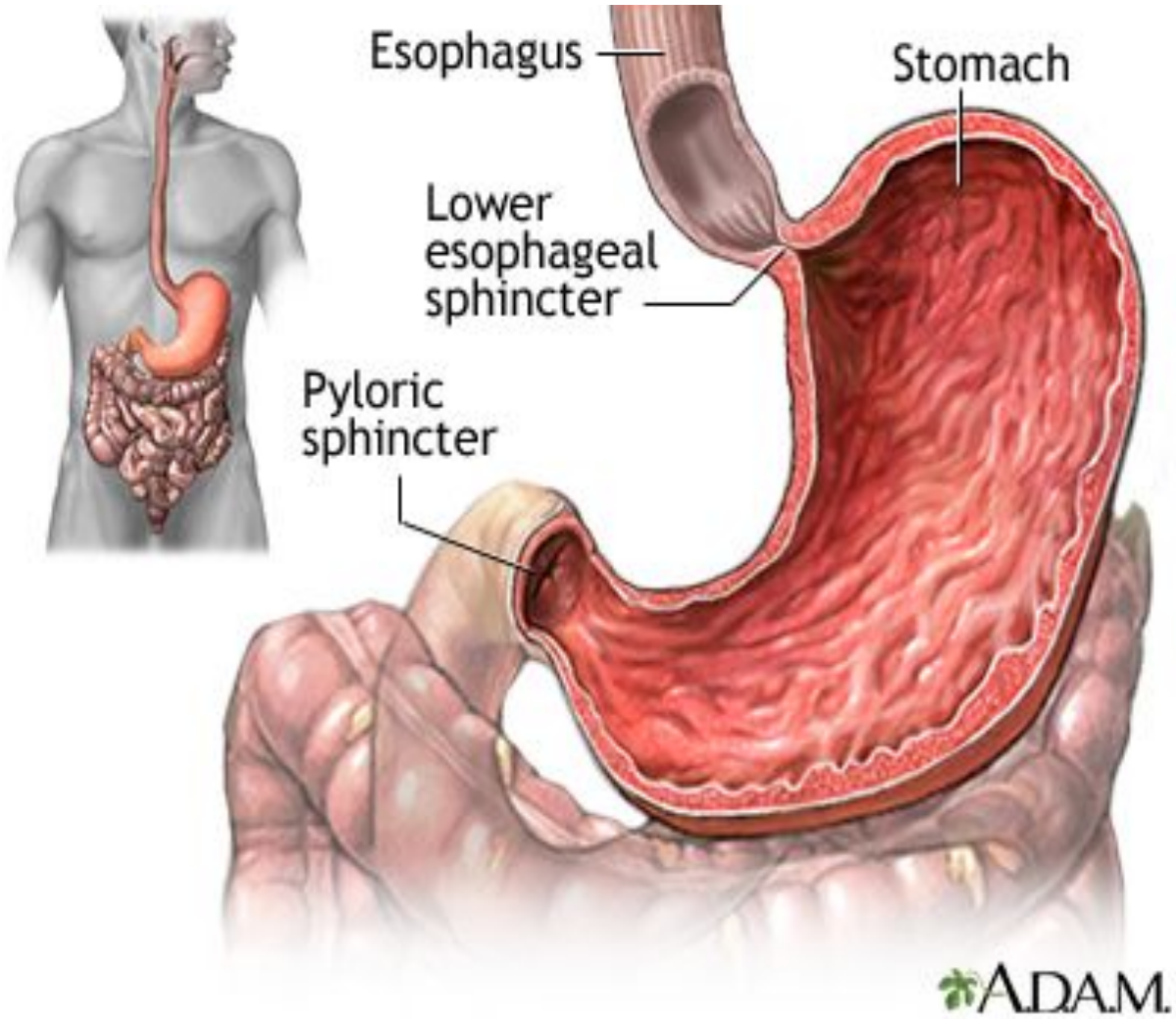
- It is a muscular tube about 25 cm/10 inches.
- It connects the pharynx with the stomach.
- It enters the abdomen passing through esophageal opening in the diaphragm at the level of T10 vertebra and opens in the stomach at the right side.



- The abdominal part of the esophagus occupies the epigastric region.
- **Blood Supply:** Left gastric artery and the veins drain into left gastric vein.
- **Nerve Supply:** Esophageal plexus via anterior and posterior vagal trunks.
- **Function:** Conducts food from the pharynx to the stomach by means of wave-like contractions called peristalsis.



- **Gastroesophageal Sphincter:** No anatomic sphincter does not exist at the lower end of esophagus, only a circular layer of smooth muscle in this part acts as a physiological sphincter.
- It’s tonic contraction prevents the gastric contents from regurgitating into the esophagus.



Stomach:

- It is the most dilated part of the GIT which is roughly 'J' shaped.
- It stores food and has a capacity of about 1500 ml.
- It mainly occupies the left hypochondriac, epigastric, and umbilical regions of the abdomen.
- It has anterior and posterior surfaces, two curvatures, the greater and lesser curvatures, and two orifices, the cardiac and pyloric orifices.
- The stomach is divided into ***four regions:***

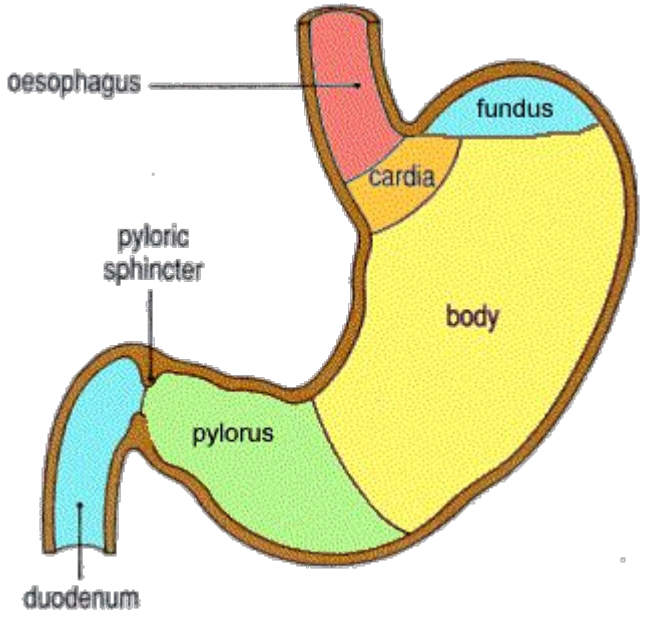
1. Cardia: surrounds the esophageal opening into the stomach.

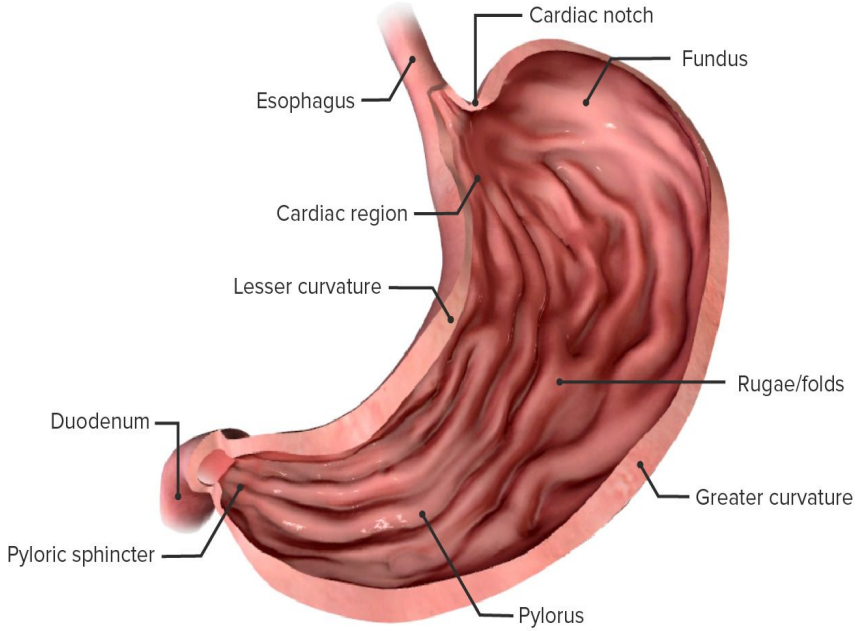
2. Fundus: dome-shaped and to the left of cardiac orifice.

3. Body: largest part of the stomach, extending from cardiac orifice to the incisura angularis

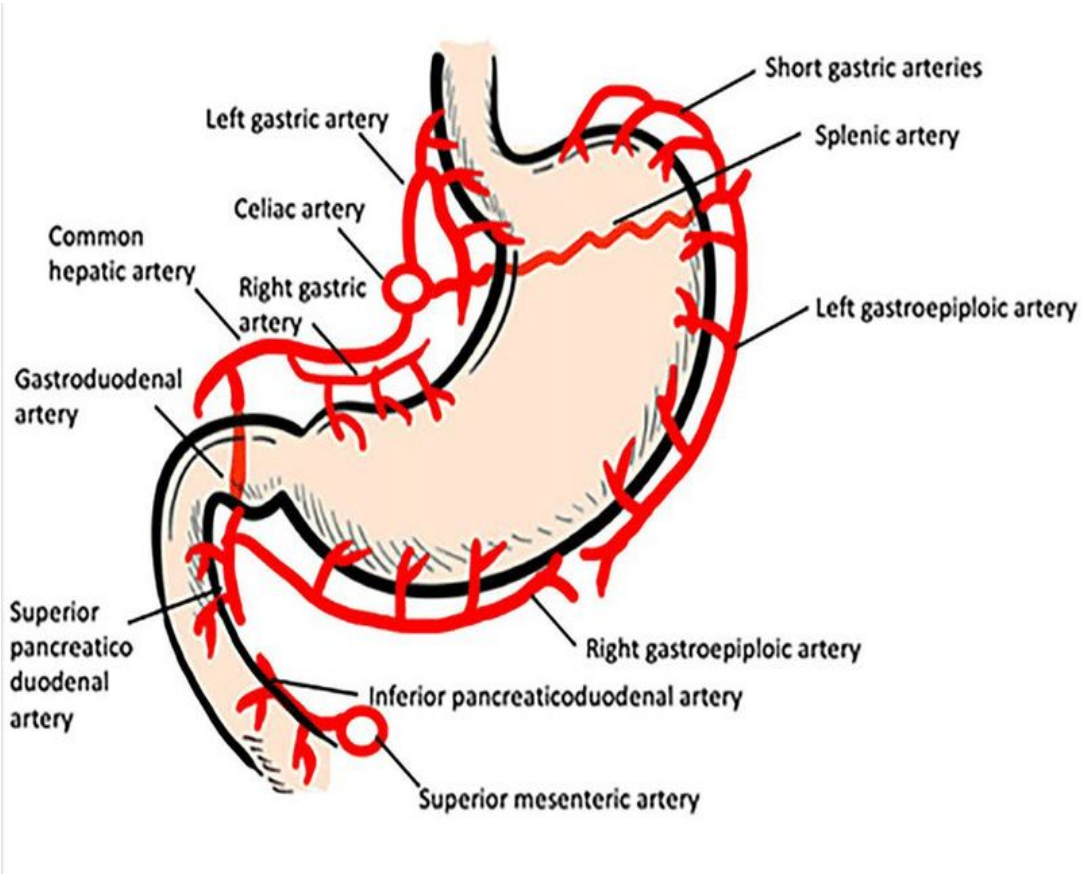
4. Pylorus: most distal part of the stomach, which contains pyloric antrum and pyloric canal guarded by pyloric sphincter.

- The pyloric sphincter controls the outflow of gastric contents into the duodenum.
- *Incisura angularis* is a constant notch on the lesser curvature.





- The mucosa of the stomach is thrown into folds called *rugae.*
- The stomach mainly occupies the epigastric region along with the left hypochondriac & umbilical regions.
- **Nerve Supply:** Sympathetic supply via coeliac plexus and parasympathetic supply via right and vagal trunks .
- **Blood Supply:** The *left gastric artery* from the coeliac trunk, descends along the lesser curvature and supplies the lower third of esophagus and upper right part of the stomach,
- The *right gastric artery* from the hepatic artery supplies the lower right part of the stomach,
- The *short gastric arteries* from the splenic artery supplies the fundus,
- The *left gastroepiploic artery* arises from the splenic artery and supplies the upper part of greater curvature, and
- The *right gastroepiploic artery*, from the gastroduodenal branch of hepatic artery supplies the lower part of greater curvature of the stomach.

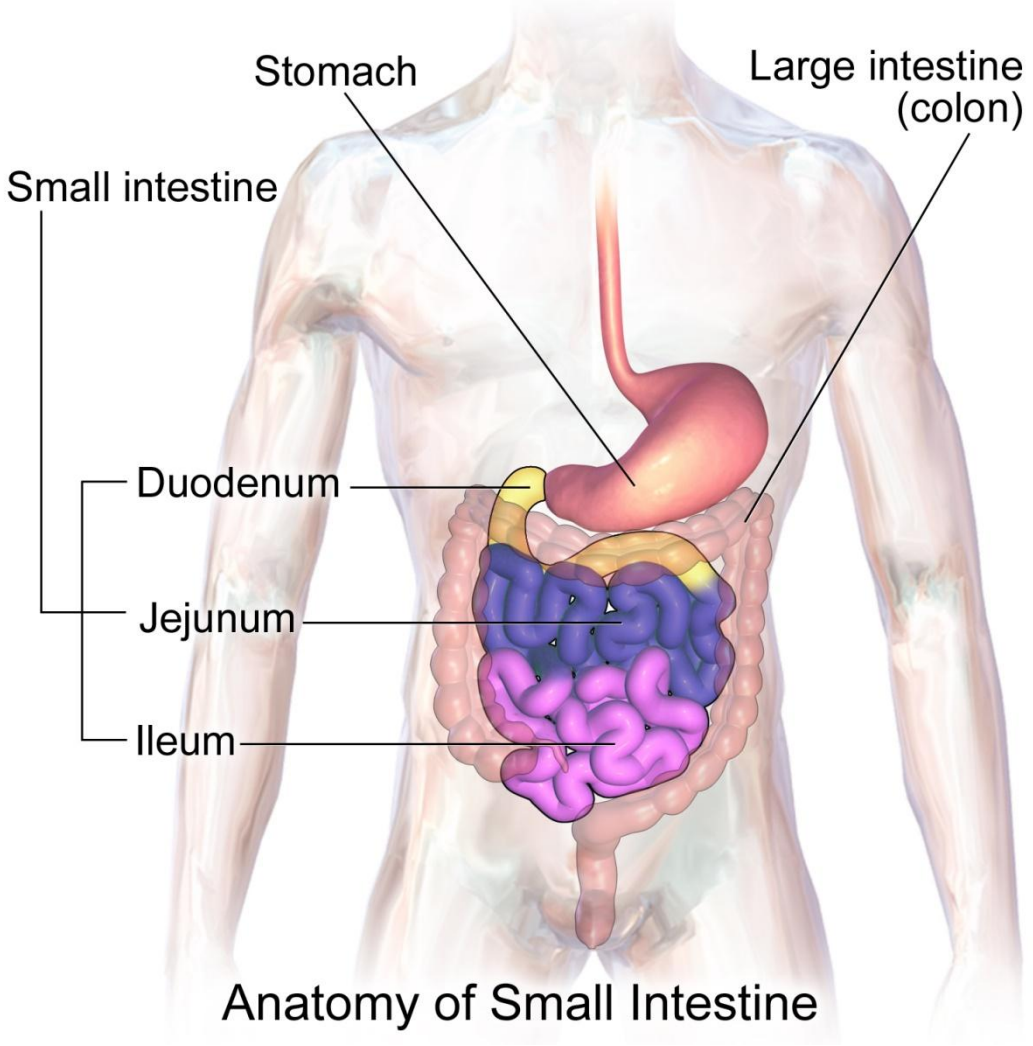


Small Intestine:

- The small intestine is the longest part of GIT.
- It extends from pyloric orifice of the stomach up to the ileo-caecal junction.
- It is approximately 6-7 meters long.
- It is divided into three parts: **duodenum, jejunum, & ileum.**

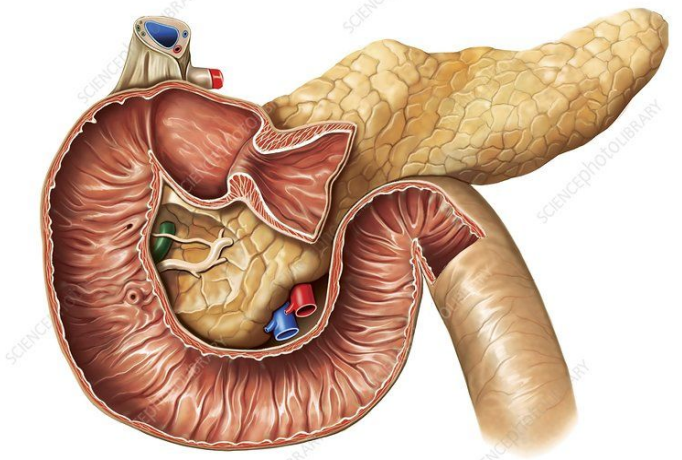
Duodenum:

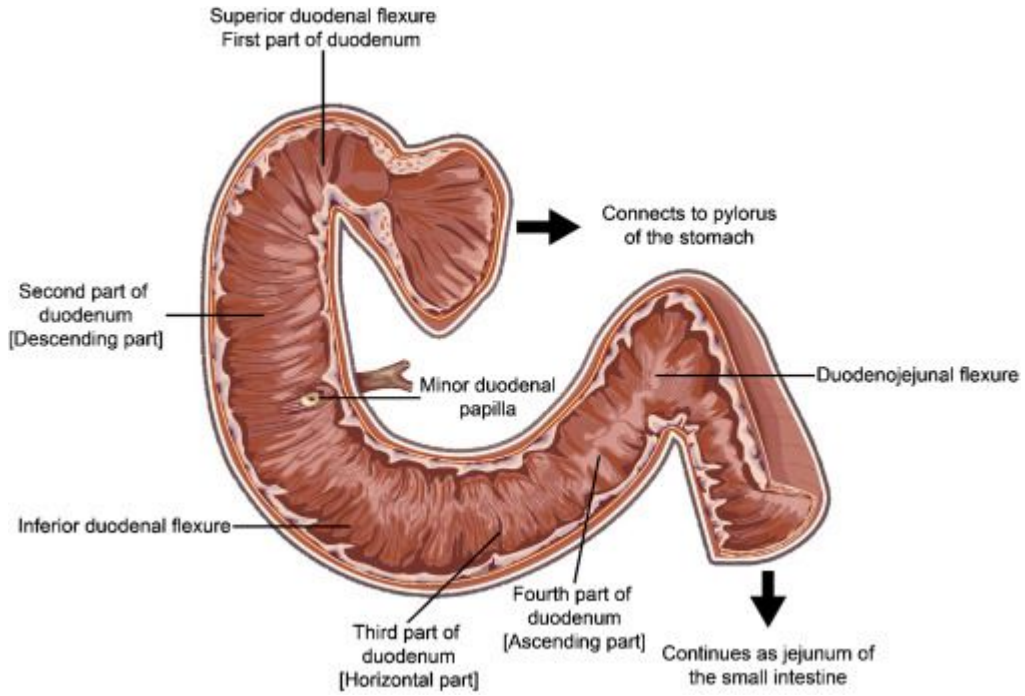
- It is the widest part of small intestine.
- It has C-shaped structure.
- It is 10 inches/25 cm in diameter.



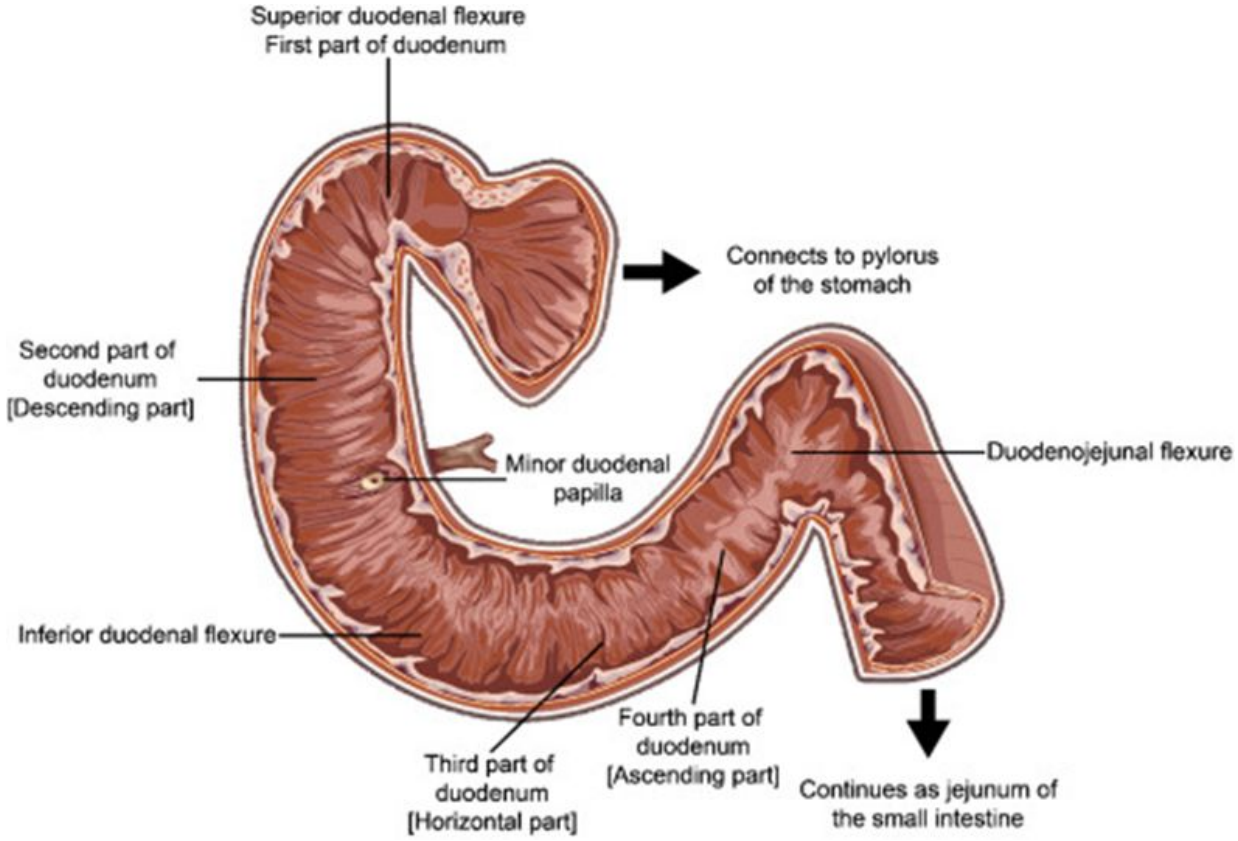
Anatomy of Small Intestine

- The duodenum is divided into four parts:
- *First/superior part:* it extends from the pylorus up to the neck of gall bladder at the level of L1 vertebra.
- Second/descending part: it runs vertically downwards on the right side to reach the L3 vertebra.
- This part of duodenum contains **major duodenal papilla** which is the common entrance for both common bile duct and major pancreatic duct, and the **minor duodenal papilla** which contains the opening for minor pancreatic duct.

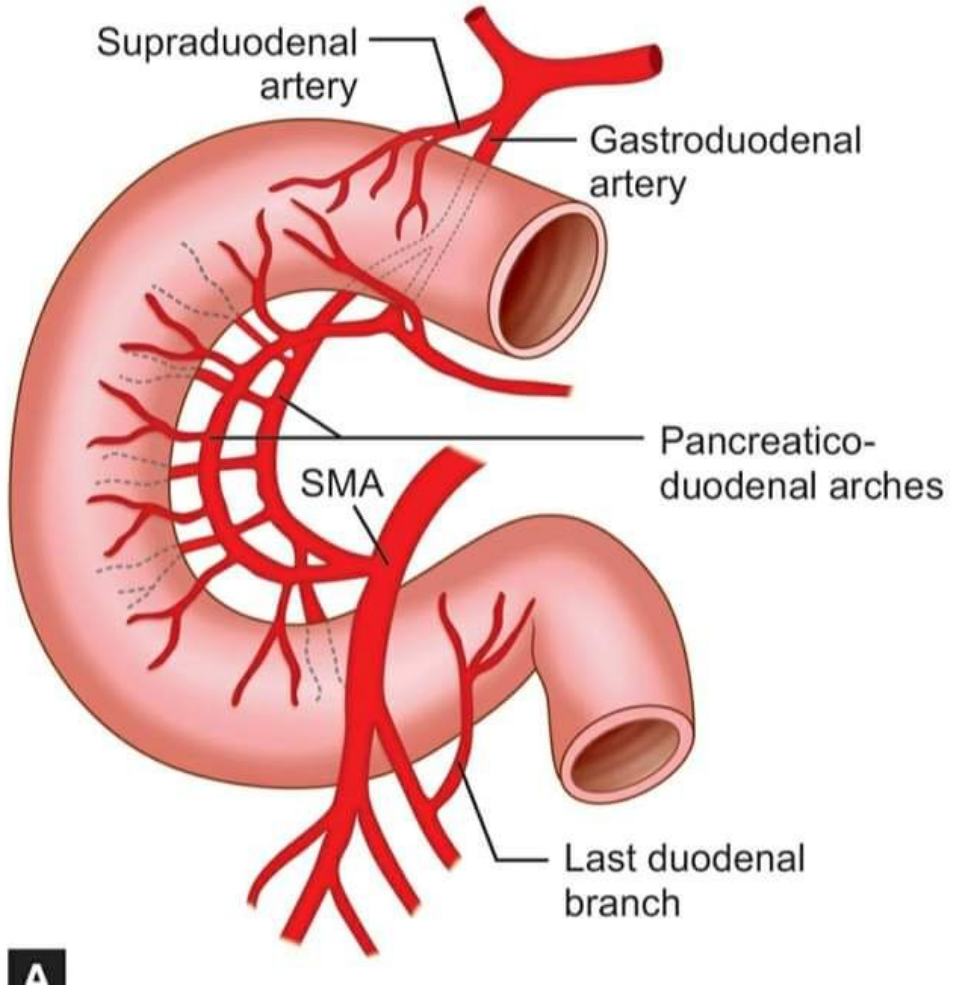




- *Third/inferior part:* it crosses the inferior vena cava, aorta, and vertebral column.
- *Fourth/ascending part:* it ascends upwards on the left side and terminates at the duodenojejunal flexure at the level of L2.
- Except for the first part, mucosa of the duodenum is thrown into numerous circular folds called **plica circulares.**

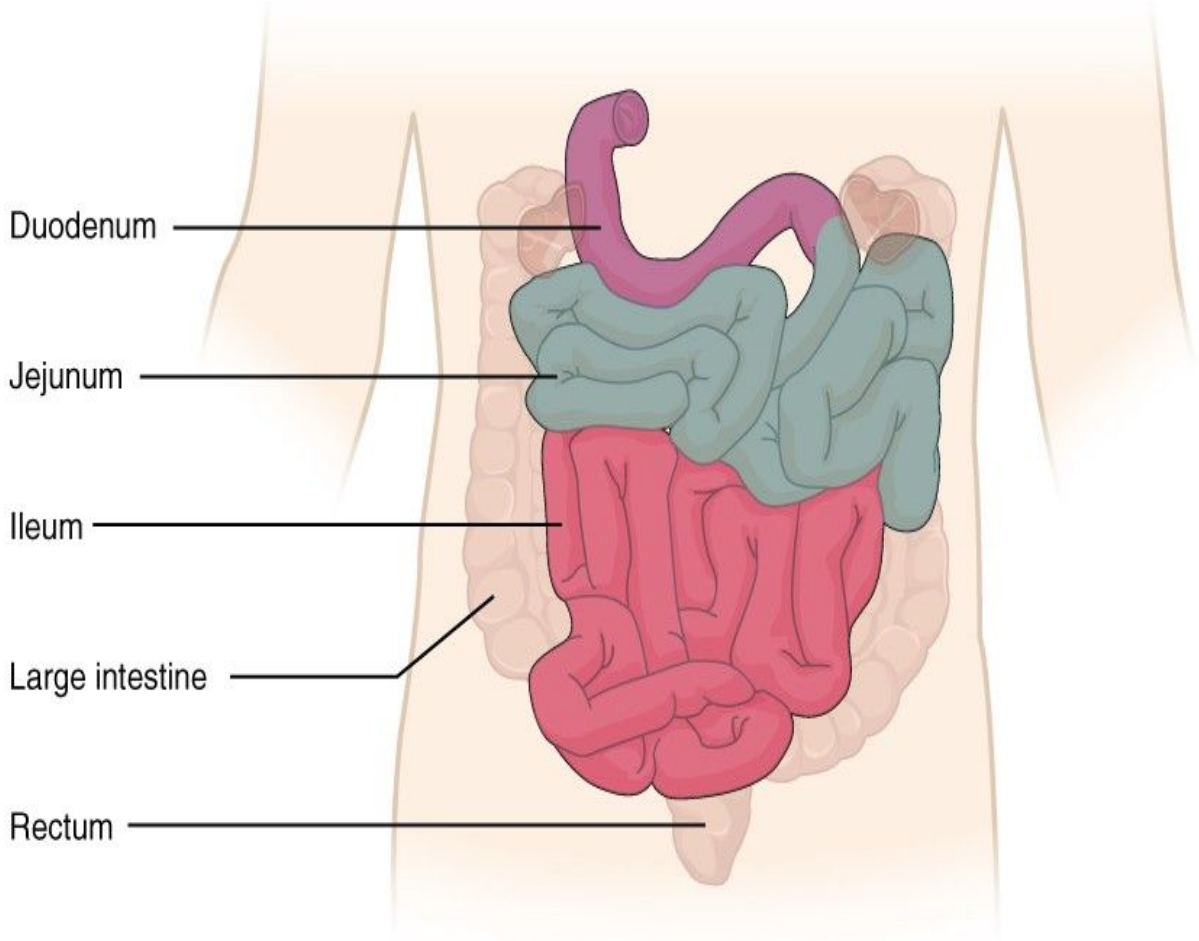


- **Blood supply :** Superior pancreaticoduodenal artery (branch of gastroduodenal artery) supplies the upper half, and
- Inferior pancreaticoduodenal artery (branch of superior mesenteric artery) supplies the lower half.
- **Nerve Supply:** parasympathetic supply via vagus nerves, and sympathetic supply via coeliac and superior mesenteric plexus.

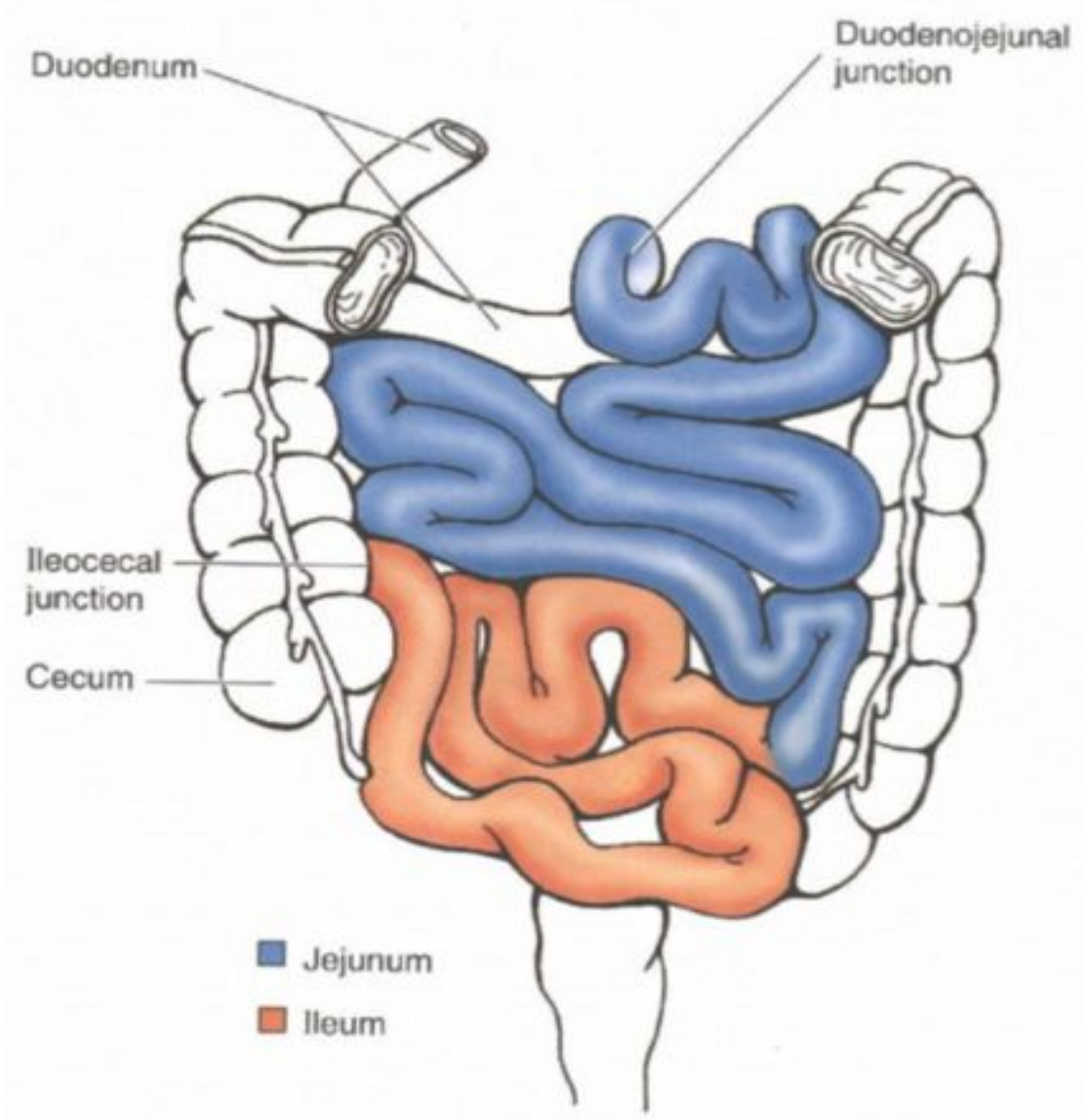


Jejunum and Ileum:

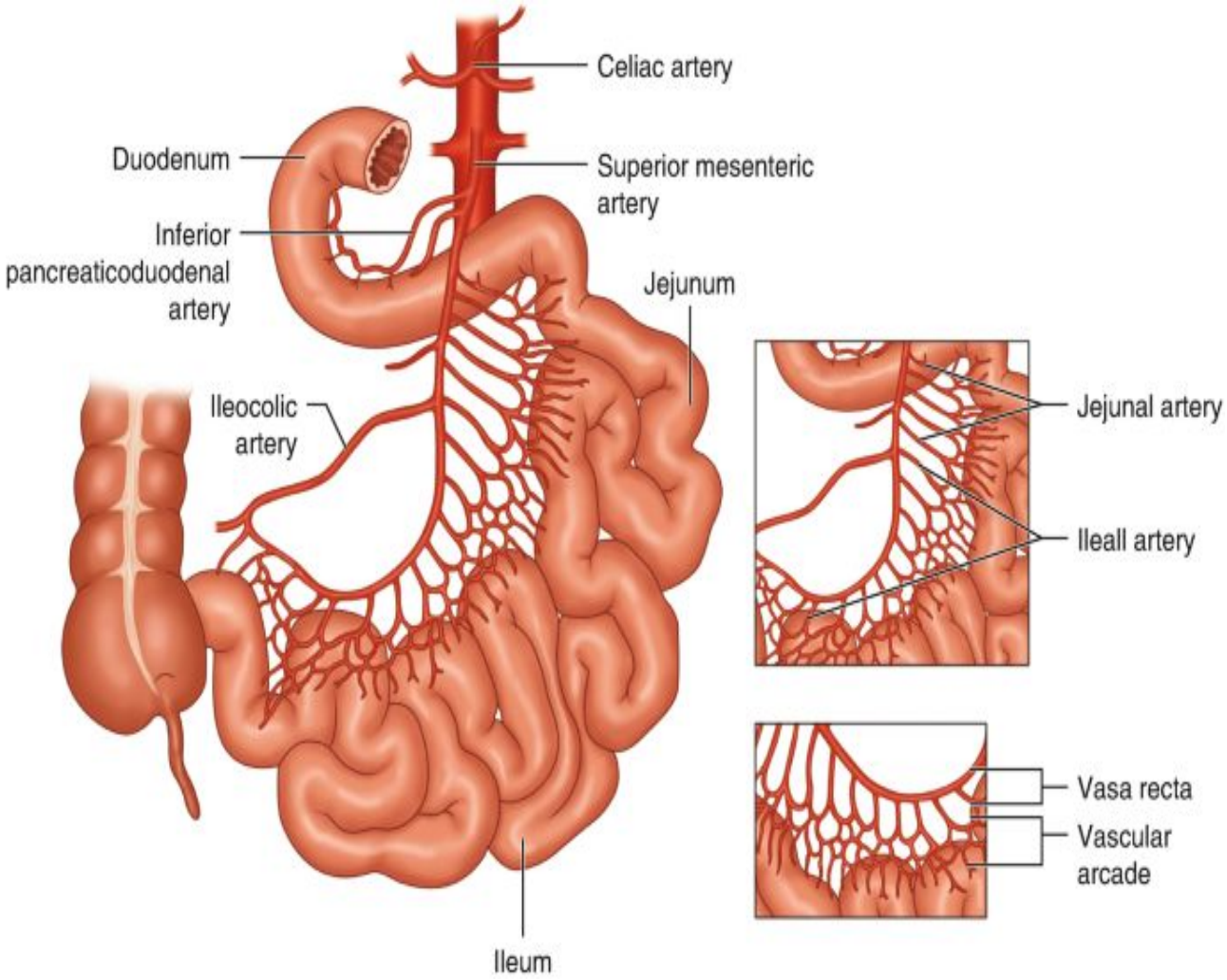
- Jejunum and ileum make up the last two parts of the small intestine.
- The **jejunum** makes up the *upper two-fifths in the left upper quadrant of abdomen*, whereas
- The **ileum** makes up the *distal/lower three-fifths in the right lower quadrant.*
- Jejunum begins at the duodenojejunal flexure and ileum ends at the ileo-caecal junction.



- The *jejunum has larger diameter with prominent mucosal folds (plica circulares)* in the lumen, whereas the *diameter of ileum is smaller, with less prominent mucosal folds*.
- *Longer vasa recta (straight arteries) and less prominent arterial arcades* are the distinctive feature of *jejunum,* whereas
- *Shorter vasa recta and more arterial arcades* are found in the *ileum.*



- **Blood Supply:** jejunal branches of superior mesenteric artery.
- The branches anastomose with one another to form a series of arch-like *arcades.*
- Multiple straight vessels (*vasa recta)* branch off the arcades and run to the wall of the intestine.
- Ileum is supplied by the ileal branches of superior mesenteric artery.

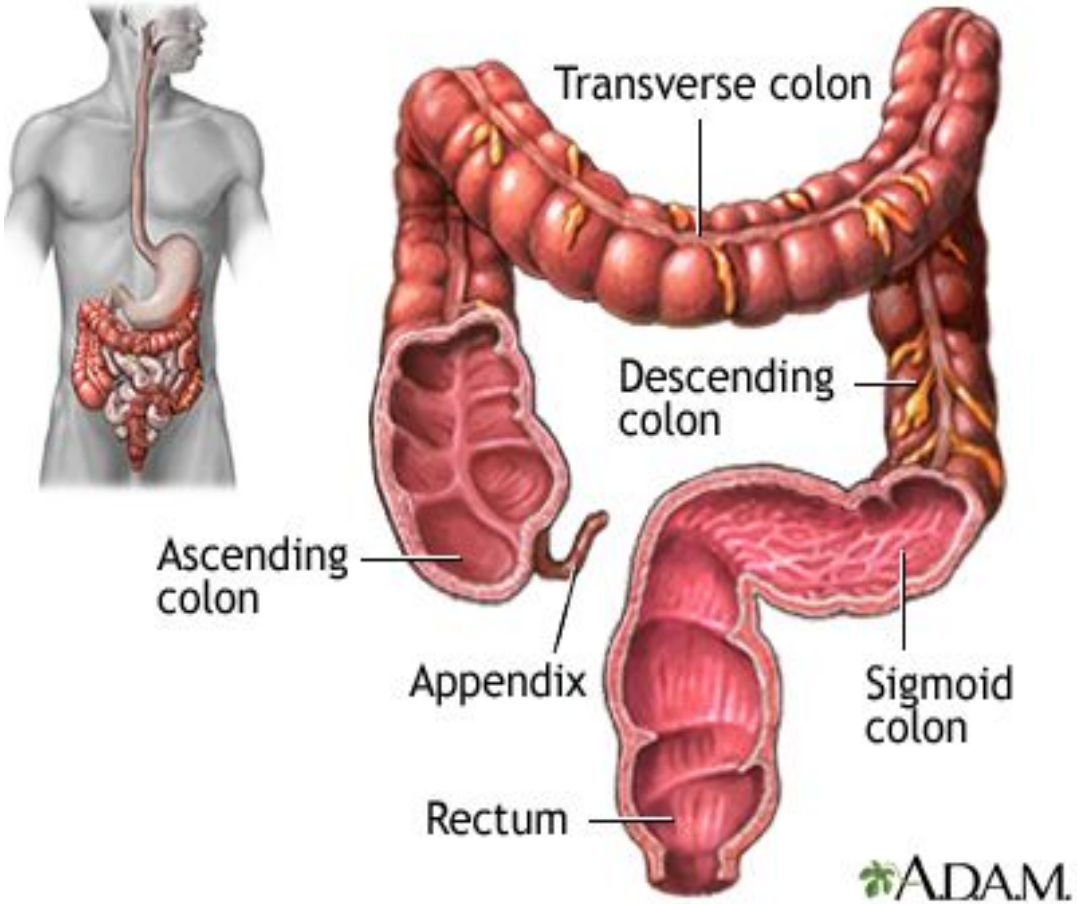


Large Intestine:

- Large intestine extends from the distal end of ileum to the anus.
- Its consists of **cecum, appendix, ascending colon, transverse colon, descending colon, sigmoid colon, rectum, and anal canal.**

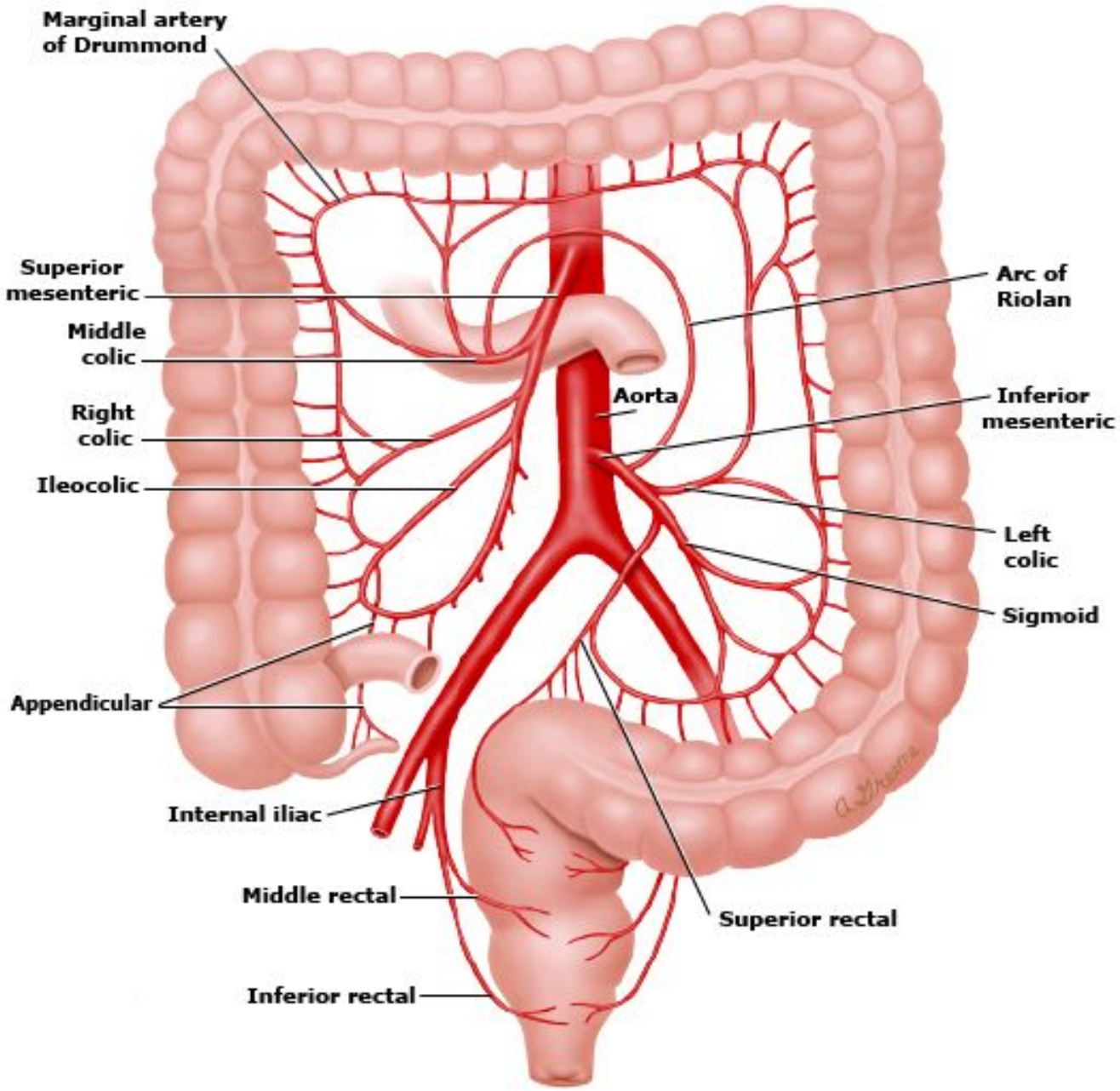
Cecum:

- It is the first part of large intestine inferior to ileocecal opening in the *right iliac fossa.*
- The ileocecal orifice contains ileocecal valve/sphincter.
- It is continuous with the ascending colon at the entrance of ileum.



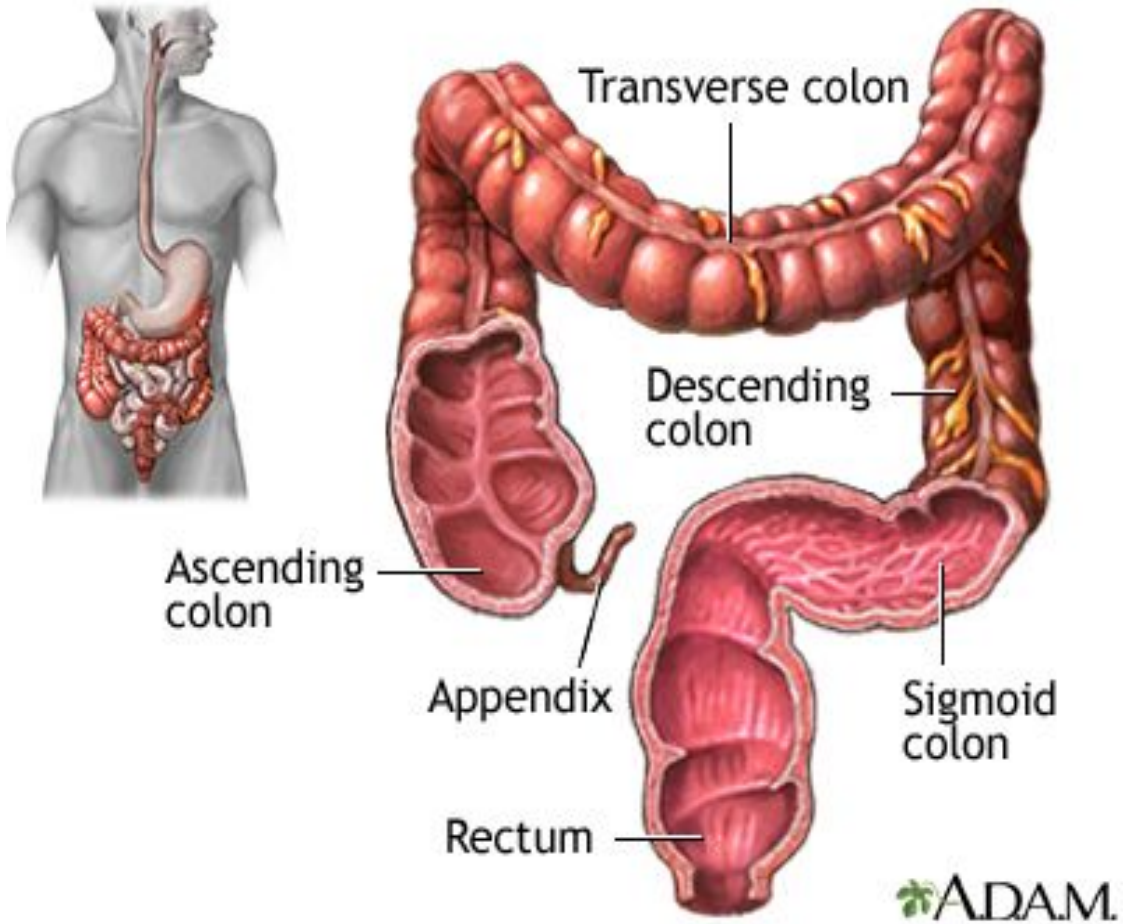
Appendix: is a narrow, hollow, blind-ended tube connected to cecum and it contains large aggregations of lymphoid tissue.

- **Arterial Supply:**
- The anterior and posterior cecal arteries, coming from ileocecal artery, and
- The appendicular artery, also arising from ileocolic artery.
- The ileocolic artery is a branch of superior mesenteric artery.



Colon:

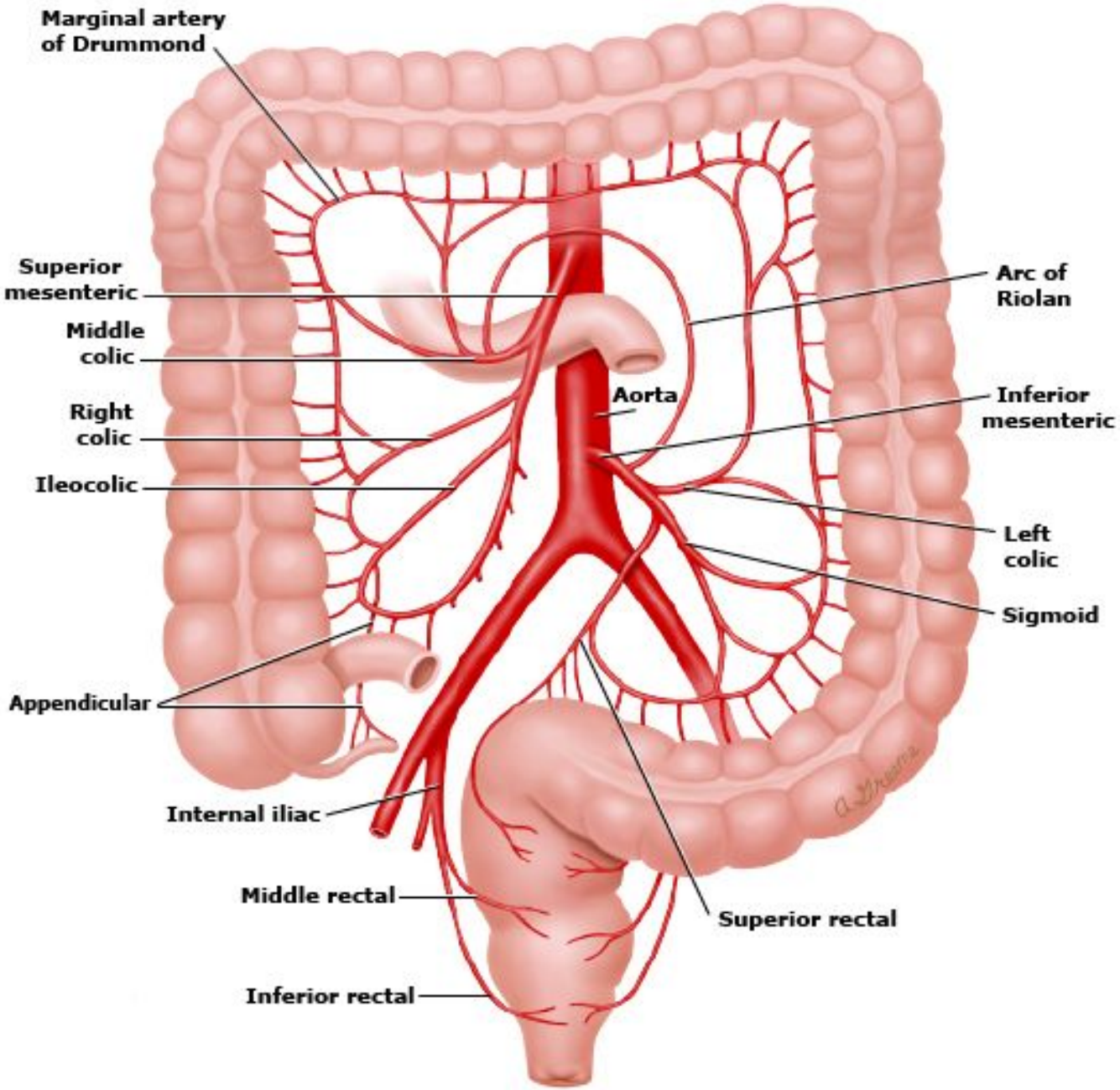
- It extends from cecum and consists of **ascending, transverse, descending, and sigmoid colon.**
- At the junction of ascending and transverse colon is the ***right colic flexure beneath the right lobe of liver.***
- Similarly, at the junction of transverse and descending colon there is another bend, the ***left colic flexure just beneath the spleen.***
- Sigmoid colon is the final segment of the colon which extends up to the level of S3 vertebra where it is continuous with the rectum.
- It is S-shaped.



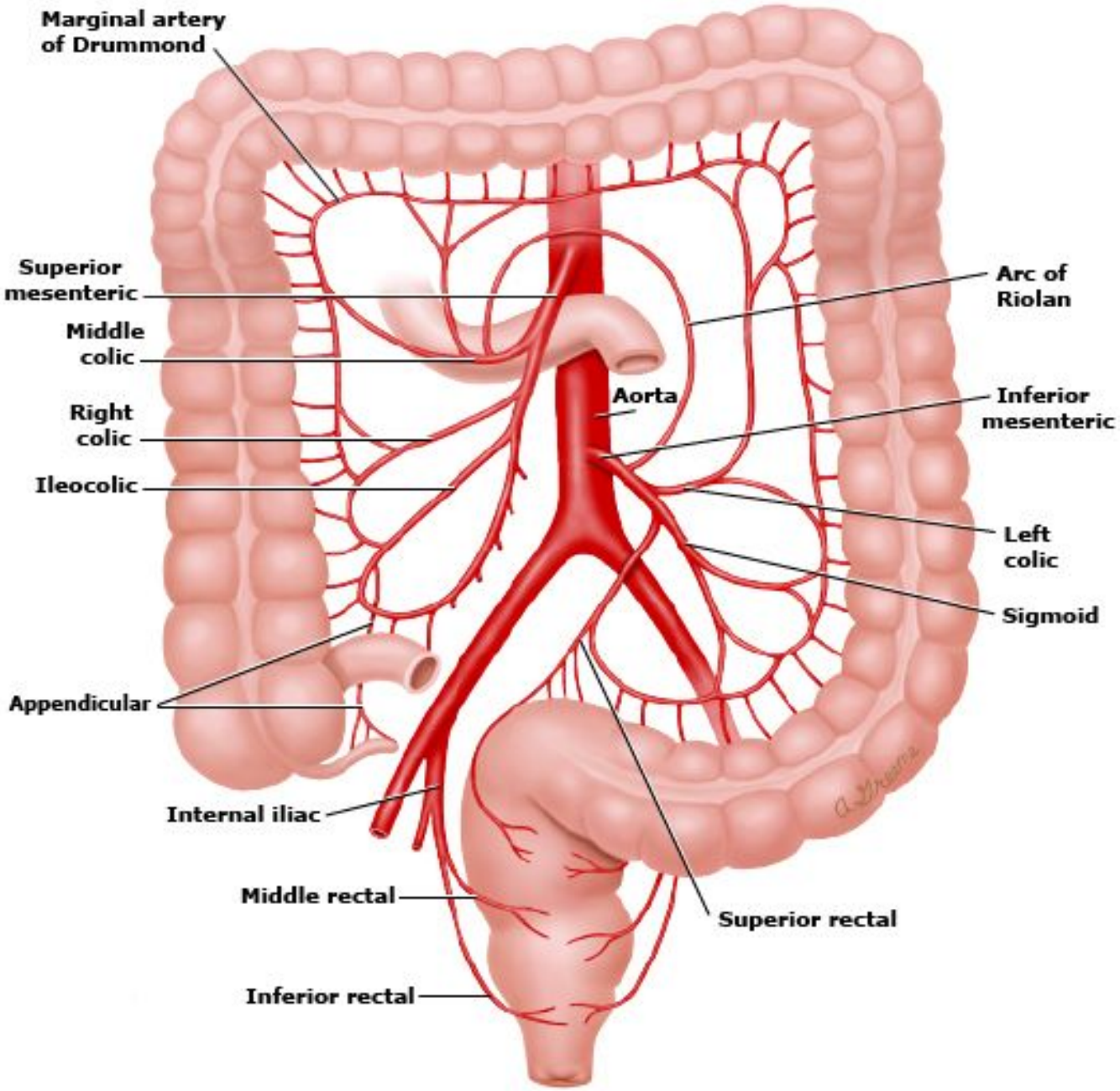
- **Arterial Supply:**
- Ileocolic, right, and middle colic branches of superior mesenteric, mesenteric artery.
- Left colic and sigmoid arteries arising from inferior mesenteric artery.

Rectum & Anal canal:

- Rectum extends from the sigmoid colon at recto-sigmoid junction.
- Recto-sigmoid junction corresponds to the level of S3 vertebra.
- Anal canal is the continuation of large intestine inferior to rectum.



- **Arterial Supply:**
- Superior rectal branch of inferior mesenteric artery,
- Middle rectal branch of internal iliac artery, and
- Inferior rectal artery arising from internal pudendal branch of internal iliac artery.

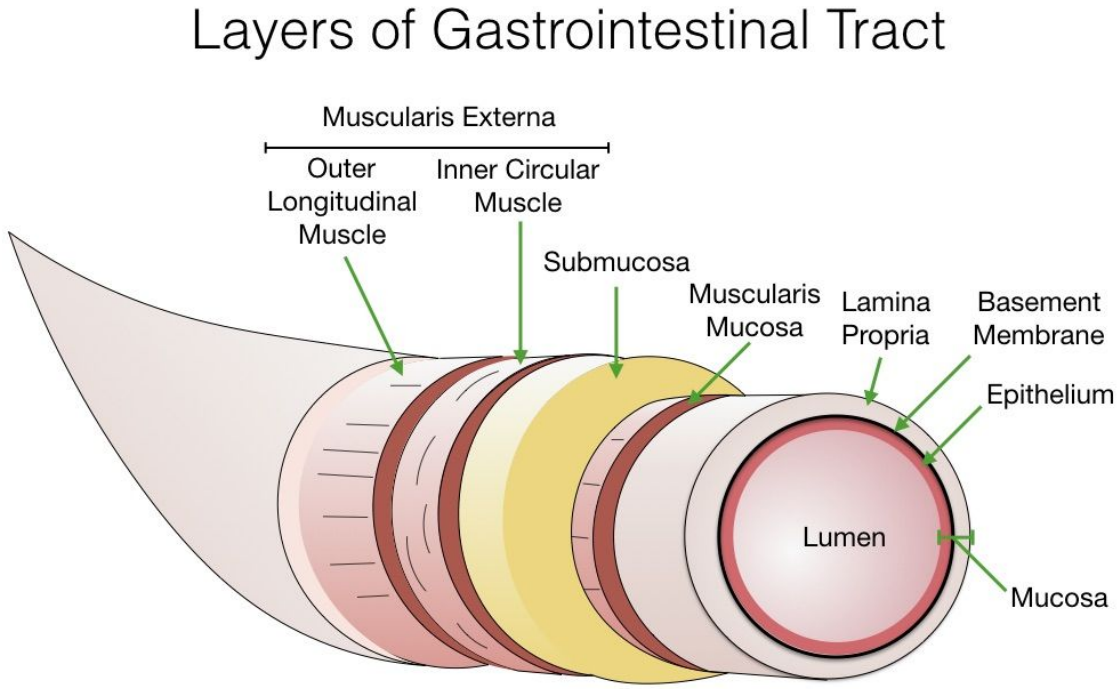


General Structural Plan of the Alimentary canal:

- All parts of the digestive tract have the same basic structure.
- Their wall consists *of four concentric layers,* the **mucosa, submucosa, muscularis externa, and serosa/adventitia.**

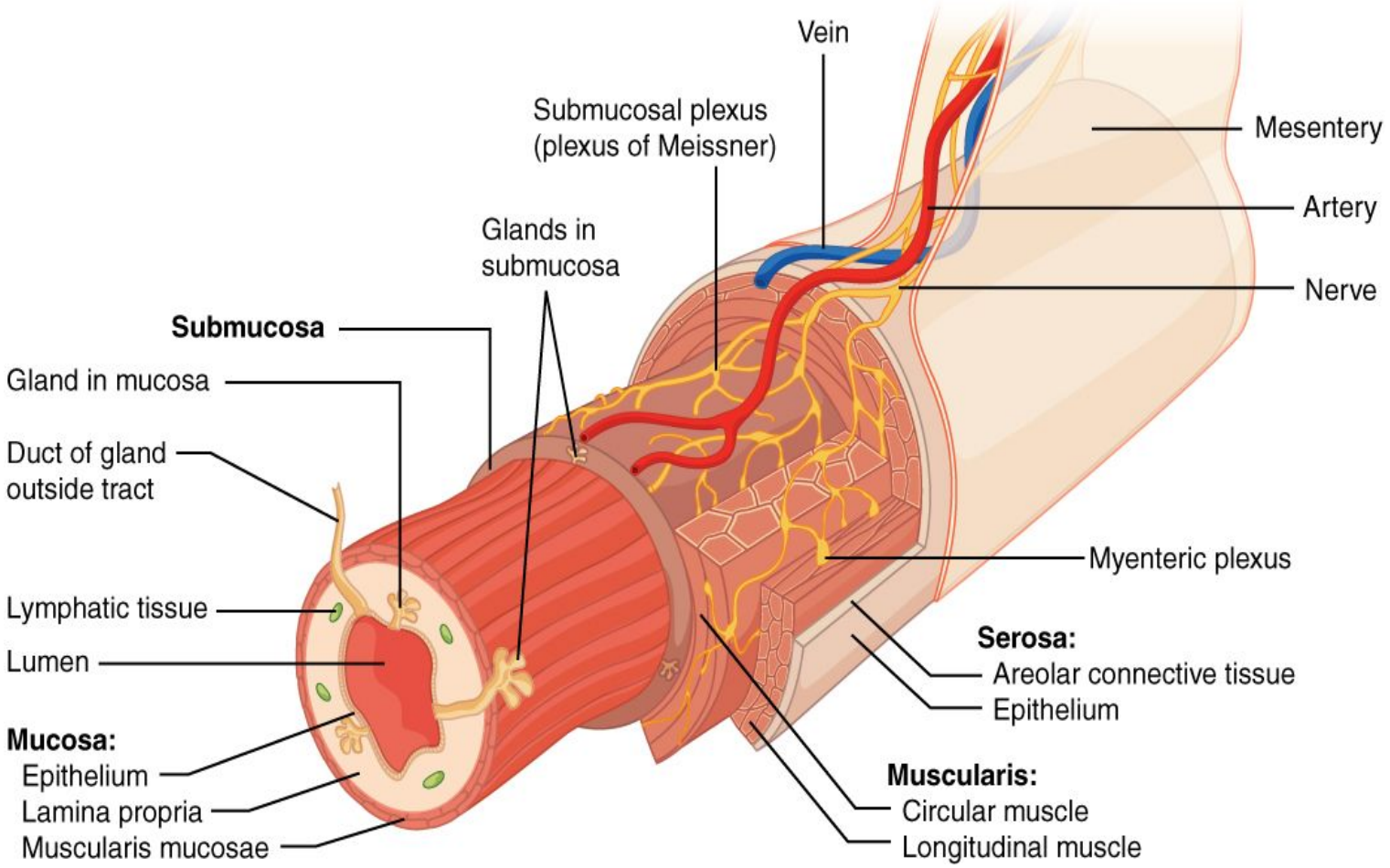
i. **Mucosa:** it consists of further 3 layers: the ***epithelium, lamina propria*** (containing mucosal glands, blood vessels, lymphatics, & MALT), & ***muscularis mucosae*** forming a boundary between mucosa and submucosa with an inner circular and outer longitudinal layer of smooth muscle.

- In ileum, large aggregates of lymphoid tissue are present in lamina propria called Peyer's patches.



ii. **Submucosa:** it is a layer of dense irregular connective tissue between mucosa and muscularis externa.

- It contains relatively larger blood vessels, lymphatics, and a nerve plexus.
- No glands are present in the submucosa except esophagus and duodenum.



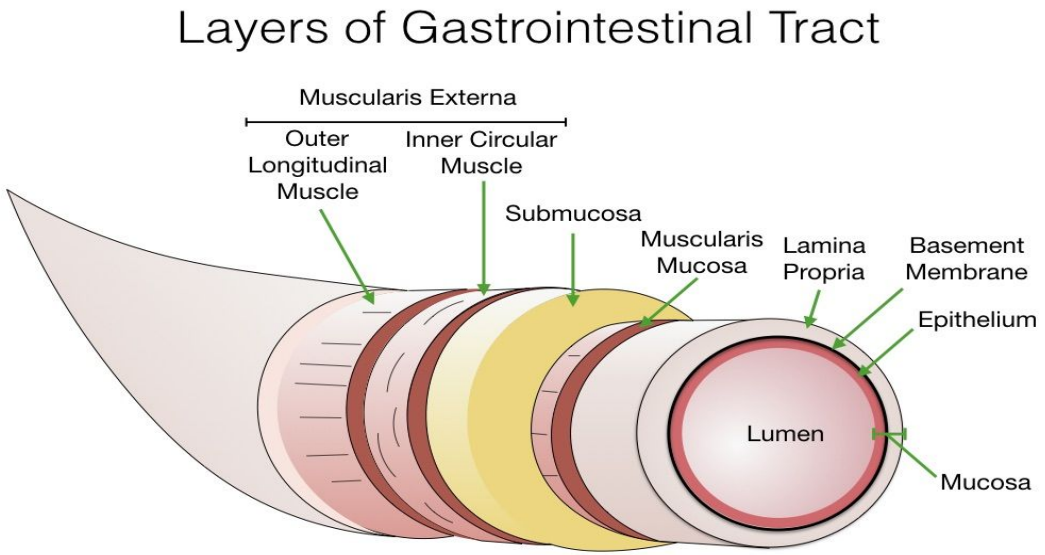
- A *submucosal/Meissner's nerve plexus is* present in the submuscosa.

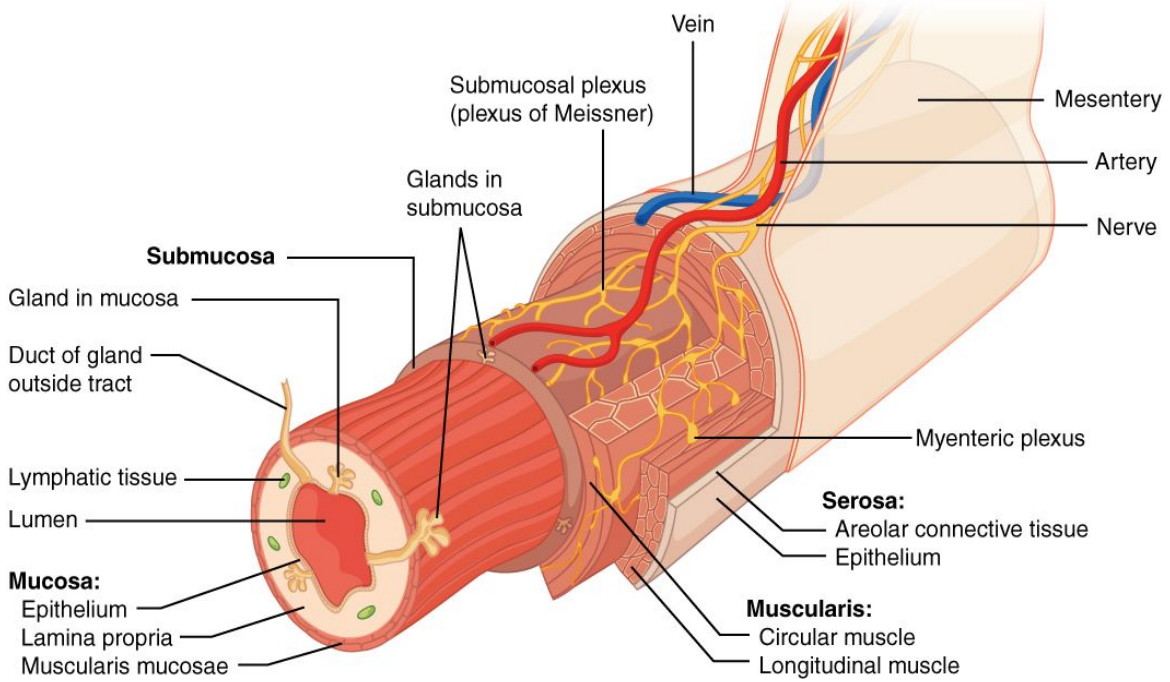
iii. **Muscularis externa:** it generally consists of two layers of smooth muscle, the inner circular & the outer longitudinal layers.

- Between the two layers is present a nerve plexus, known as the *myenteric/Auerbach plexus.*

iv. **Adventitia/serosa**

- **N.B: esophagus** is lined by **stratified squamous non-keratinized** epithelium, GIT from **stomach up to the upper part of anal canal** is lined by **simple columnar epithelium**, lower part of anal canal is lined by stratified squamous non-keratinized epithelium.

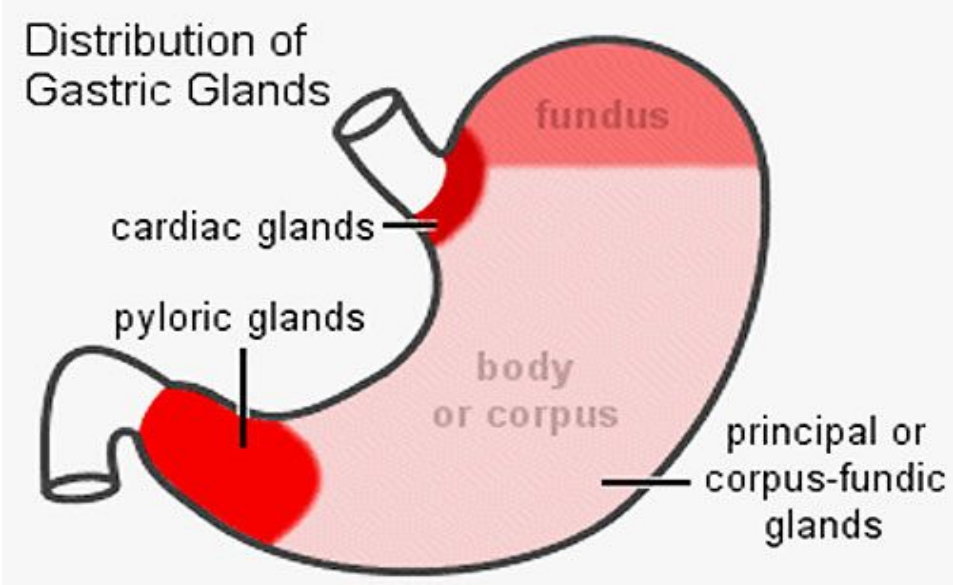


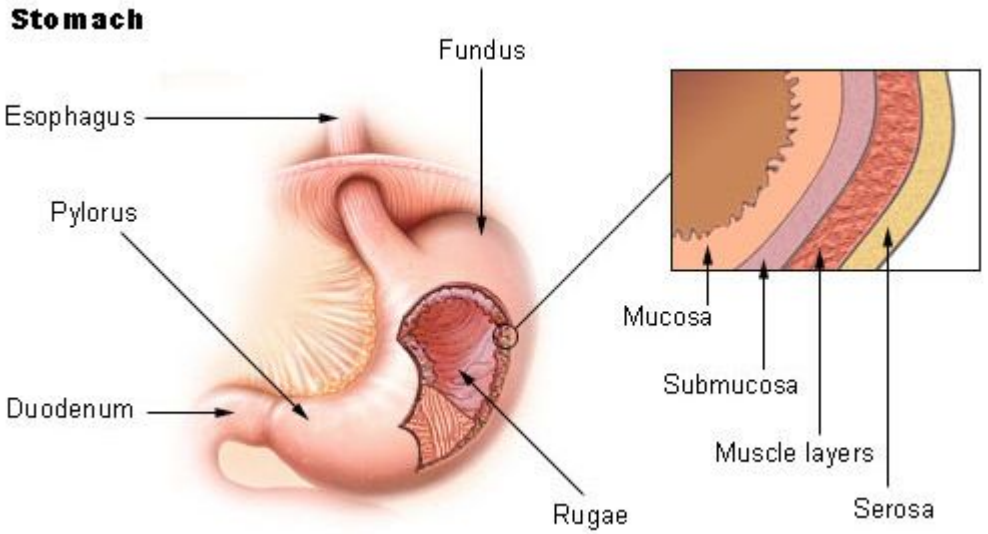


Glands of the Stomach:

- The lamina propria of gastric mucosa contains numerous glands which can be classified into 3 types on the basis of their distribution & structure:

1) **Gastric glands:** found in the fundus & body of the stomach.
2) **Cardial glands:** lie around the inlet of the stomach.
3) **Pyloric glands:** located in the pyloric antrum & pyloric canal.

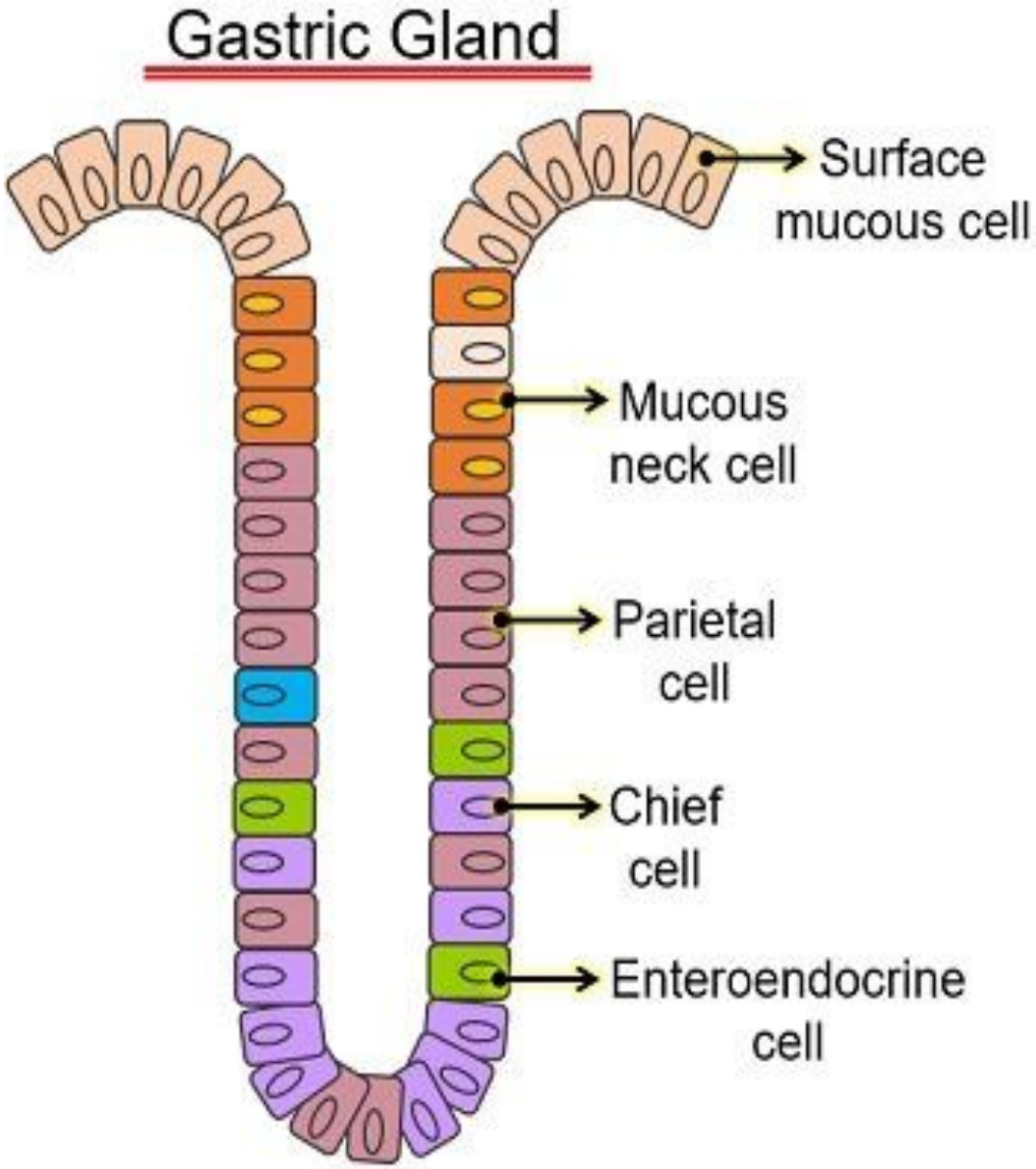




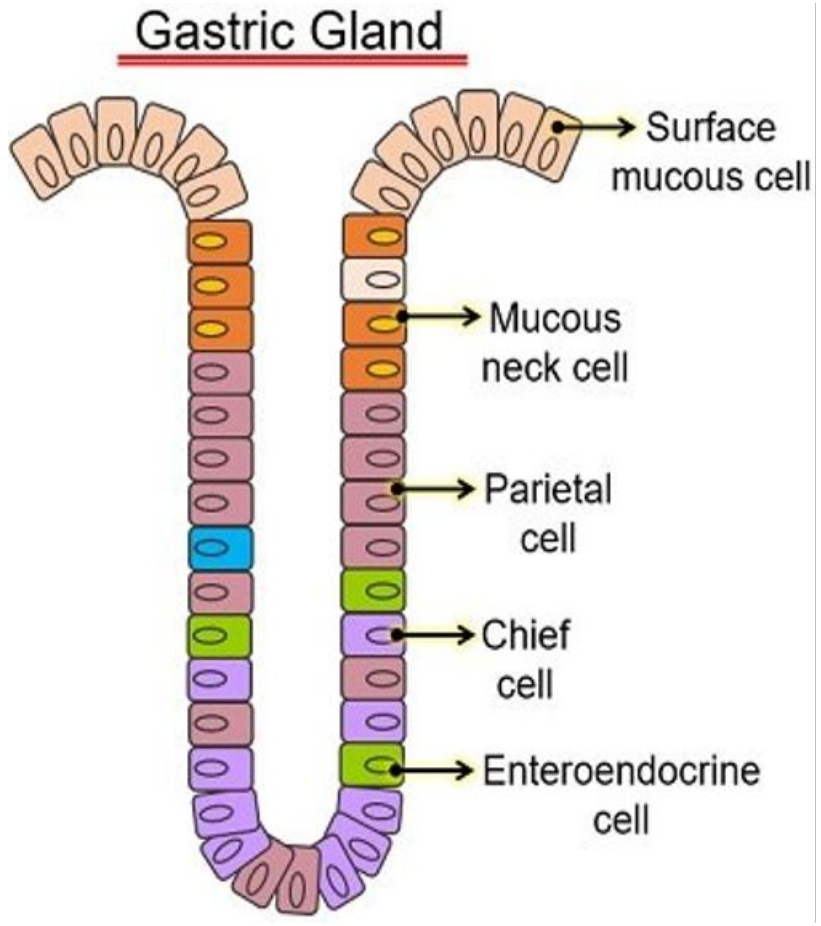
1) GASTRIC GLANDS:

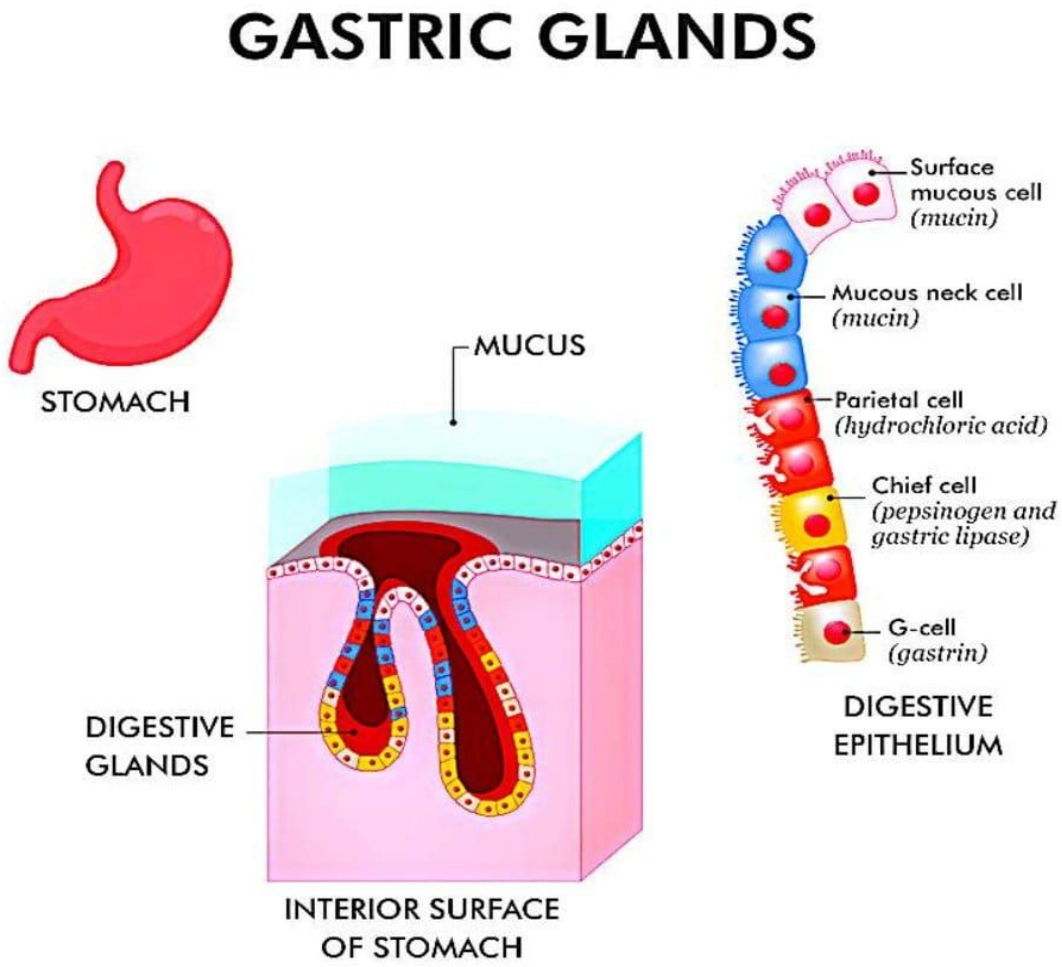
- Also known as *fundic glands.*
- They are of **simple branched tubular** type.
- They have isthmus, neck & base.
- They mainly produce gastric juice, acid & most of the enzymes secreted by the stomach.
- Their lining epithelium contains 5 types of cells:

i. Mucus neck cells
ii. Parietal cells
iii. Chief cells
iv. Enteroendocrine cells
v. Stem cells



Cells of Gastric Glands with their function:



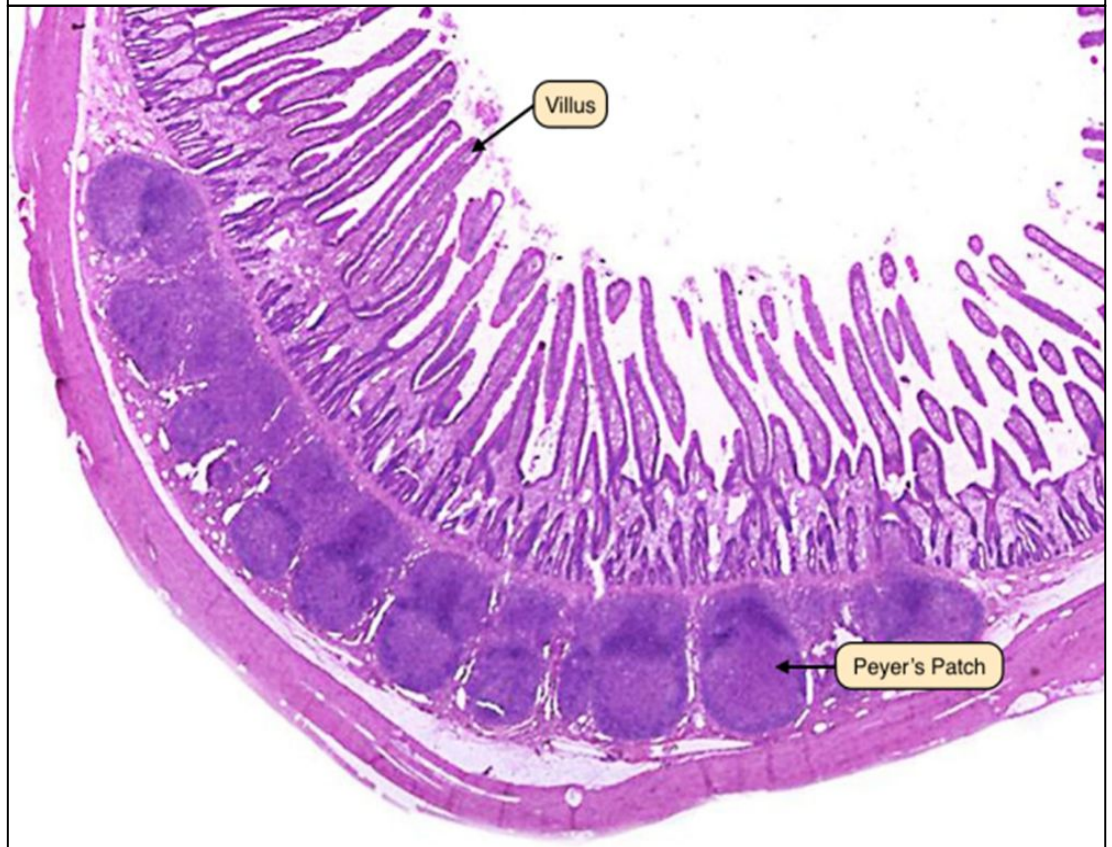


General differences between small & Large Intestines:

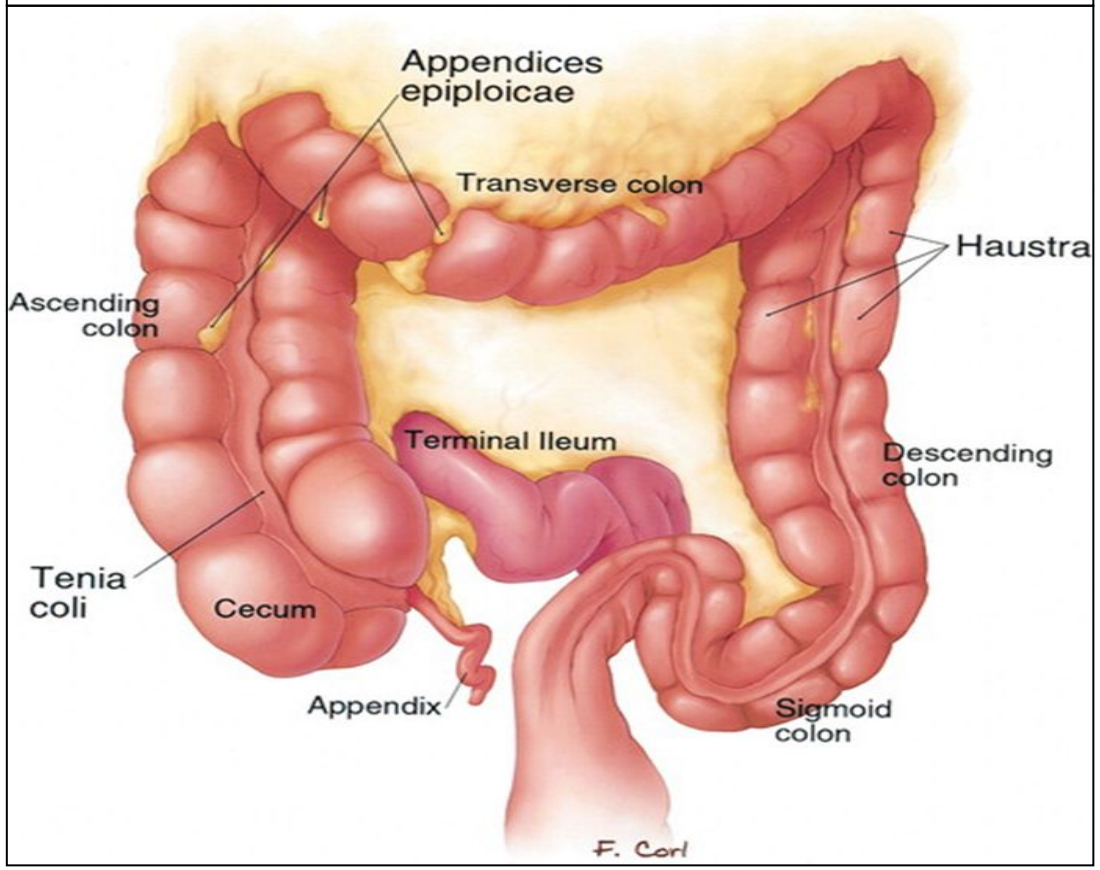
Small Intestine	Large Intestine
1) The diameter of full small intestine is smaller.	1) The diameter of full large intestine is smaller.
2) The longitudinal muscle coat in small intestine forms a continuous coat around the gut.	2) Whereas the longitudinal muscle coat in large intestine except appendix, is collected into three bands called **teniae coli.**
3) The small intestine has no fatty tags attached to its wall.	3) The large intestine has fatty tags called **appendices epiplocae/ epiploic appendages.**
4) The wall of small intestine is smooth.	4) The wall is sacculated in large folds called **haustrations/ haustra.**

Features specific to small/large intestine:

Small intestine (ileum):



Large intestine:



Cells of Intestinal Epithelium with Function:

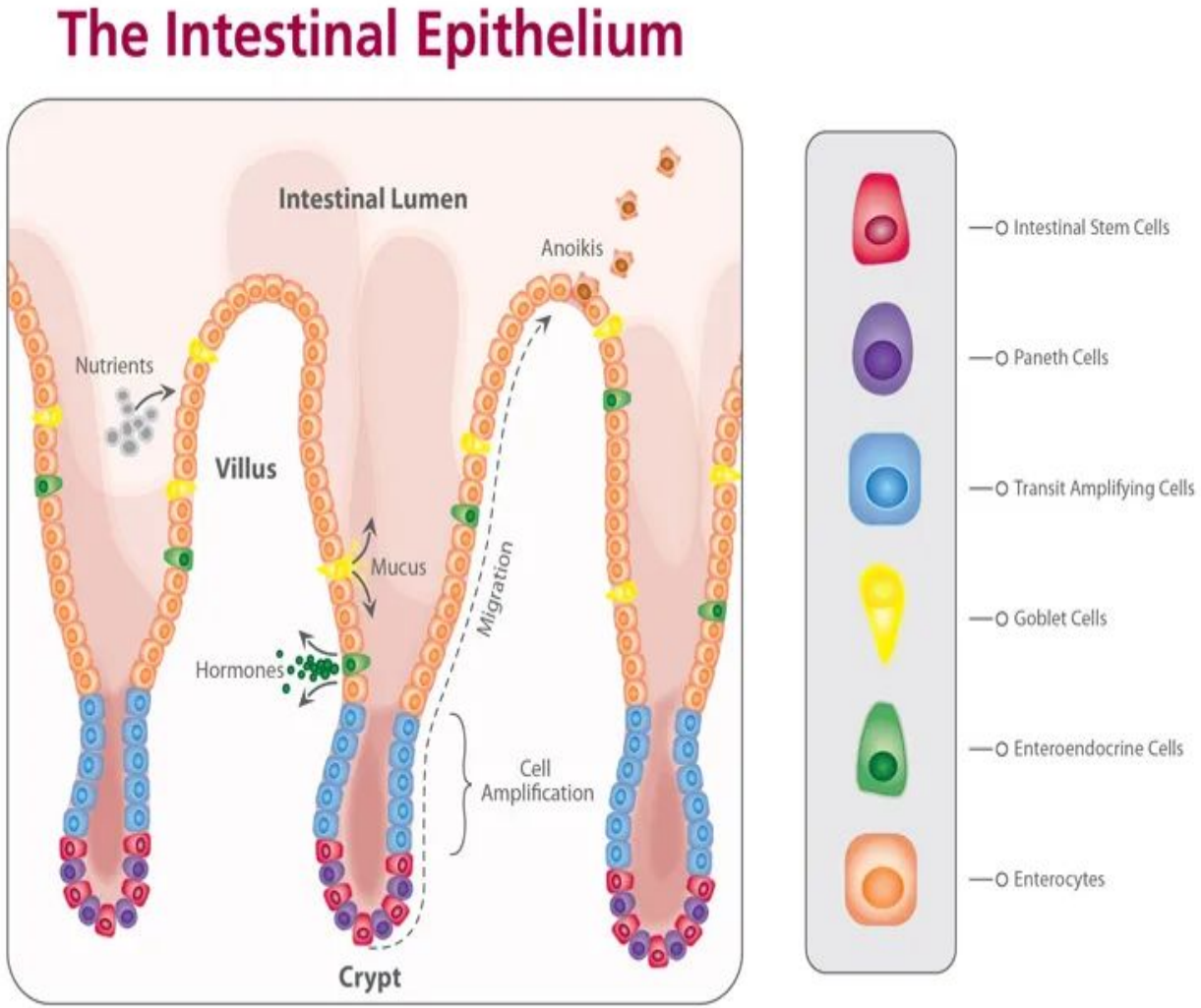


Table 2
Intestinal epithelial cell types and function

Cell	Function	Structure	Location
Enterocytes	The main absorptive cell of the small intestine	Tall columnar cells with microvilli on the apical side; bound to each other by junction complexes	Predominant cell type of the epithelium
Paneth cells	Secrete lyzosyme and defensing, which serve as a defense against bacteria and may regulate intestinal flora	Large, pyramidal cells with eosinophilic granules at their apical end that contain lysozyme and other glycoproteins	Base of crypts of Lieberkuhn
Enteroendocrine cells	Locally regulate intestinal activity by secreting hormones (CCK, secretin, GIP) into the capillaries	Granules are closer at the basal rather than apical end of the cell	Primarily located in the crypts but can be found at other locations within the epithelium
Goblet cells	Mucin secreting cells that serve a cytoprotective role and lubricating role in the GI tract	Goblet shaped with large granules located apically in the cell	Located throughout the small intestine, greater concentration in the ileum than the jejunum
Tuft cells	Serve a role in chemoreception and potentially parasitic defense	Have an apical bundle of microfilaments connected to a long microvilli that protrudes into the lumen	A rare cell type comprising 0.4% of intestinal epithelium
Cup cells	The apical end is broad and tapers toward the base	No current known function	Limited to the ileum
M cells	Antigen sampling cells that endocytose bacteria and viruses from the lumen to Peyer patch	Epithelial cells that lack a brush border and have many apical endocytic vesicles	Overlie Peyer patch

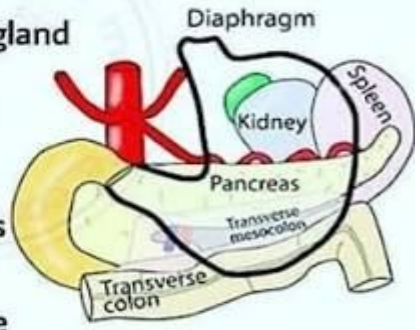
The Stomach Bed:

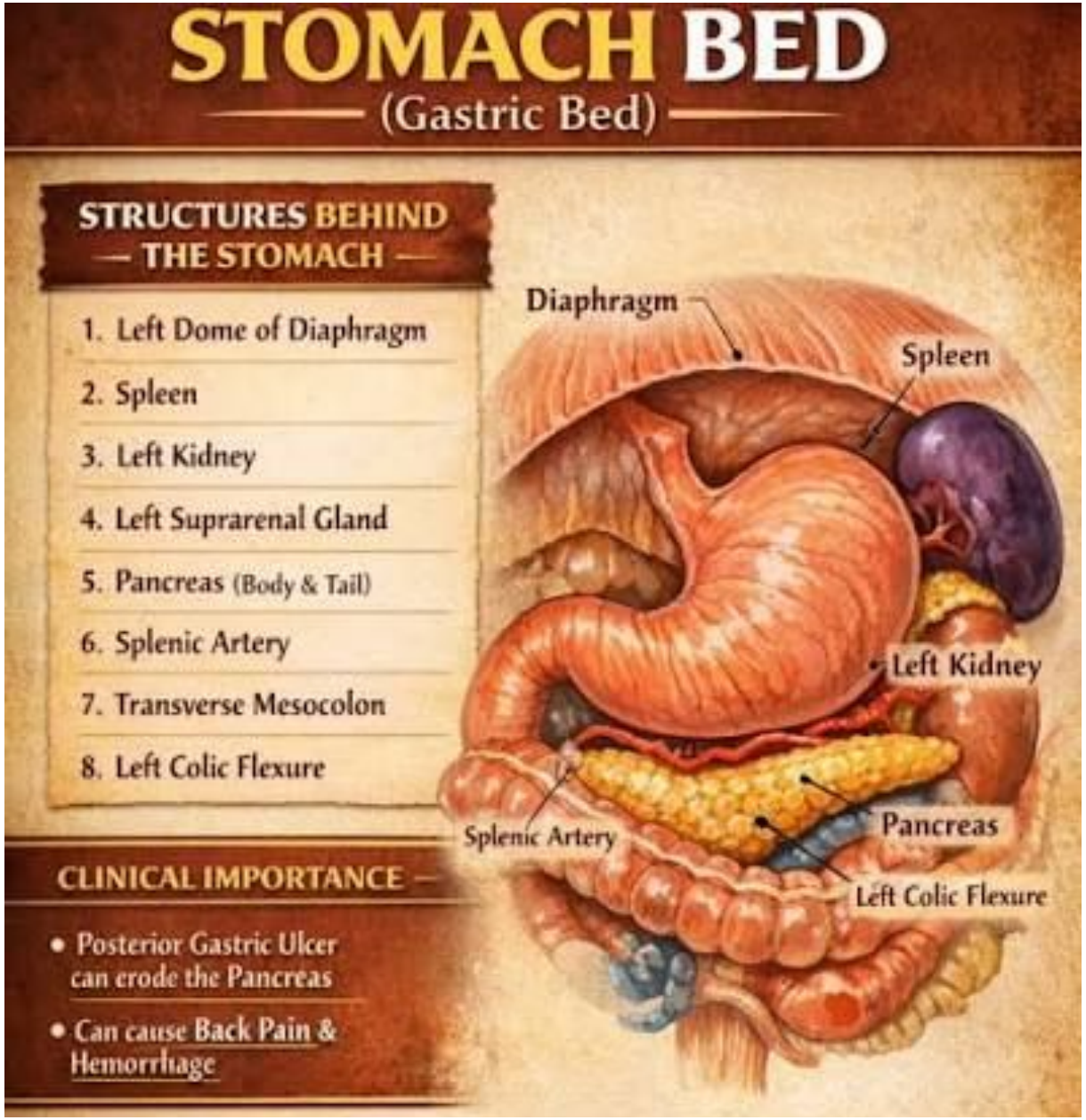
- The structures on which the posterior surface of stomach rests, form the ***'Stomach Bed'***.

Stomach Bed

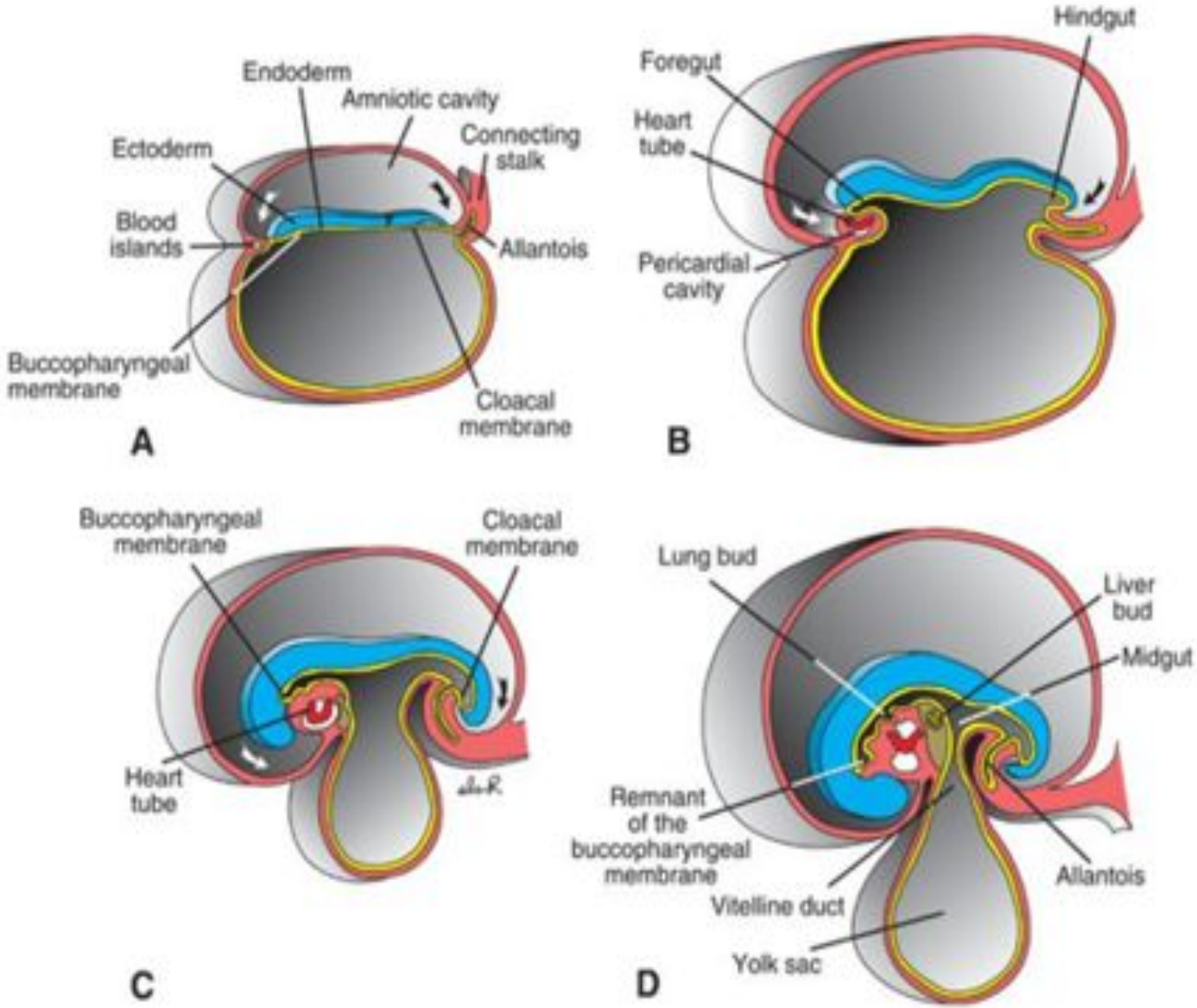
Dr. S S Kills Patient cruelly Mercilessly

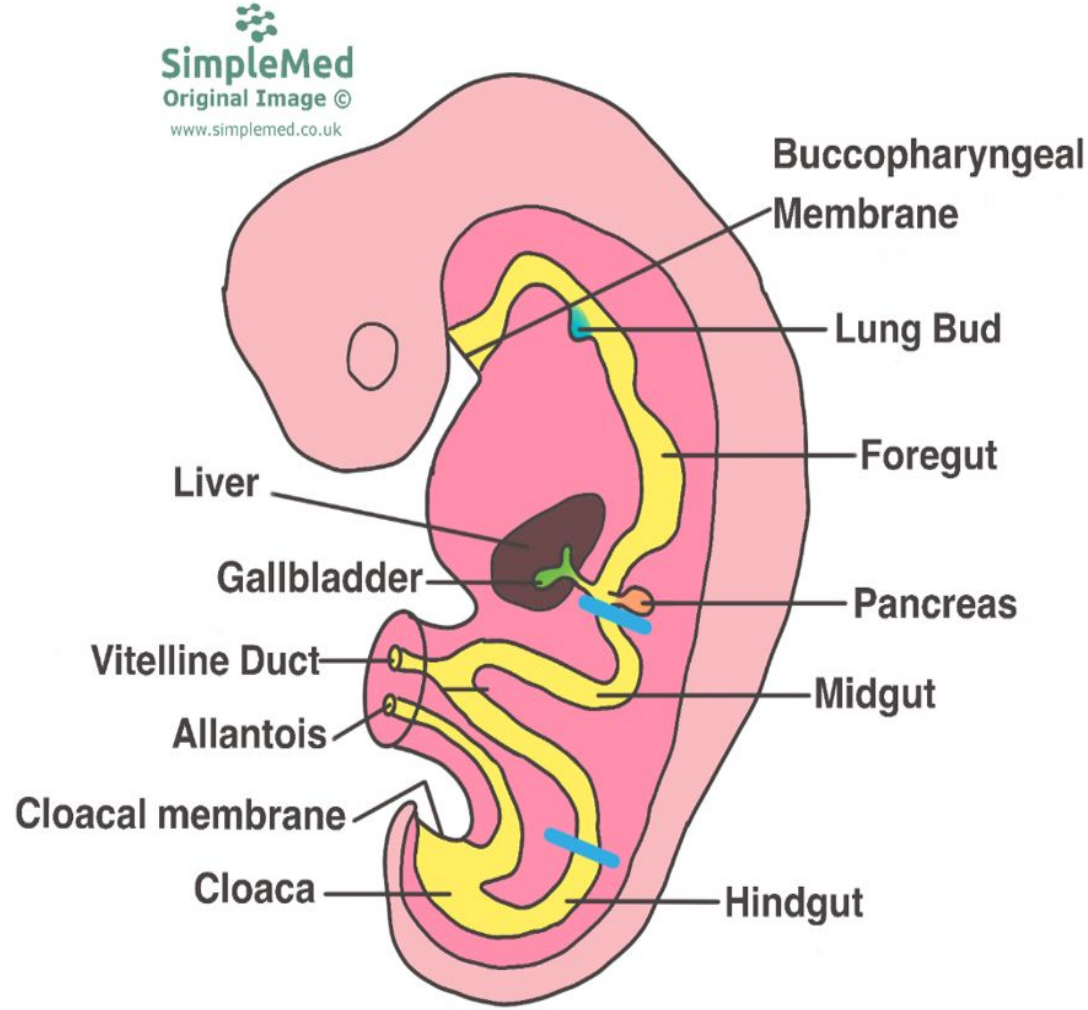
Dr.	- Left crus of Diaphragm
S	- Splenic artery
S	- Left Suprarenal gland
Kills	- Left Kidney
Patient	- Body of Pancreas
Cruelly	- Left Colic flexure
Mercilessly	- Transverse Mesocolon





RECALL: Incorporation of Primitive Gut Tube



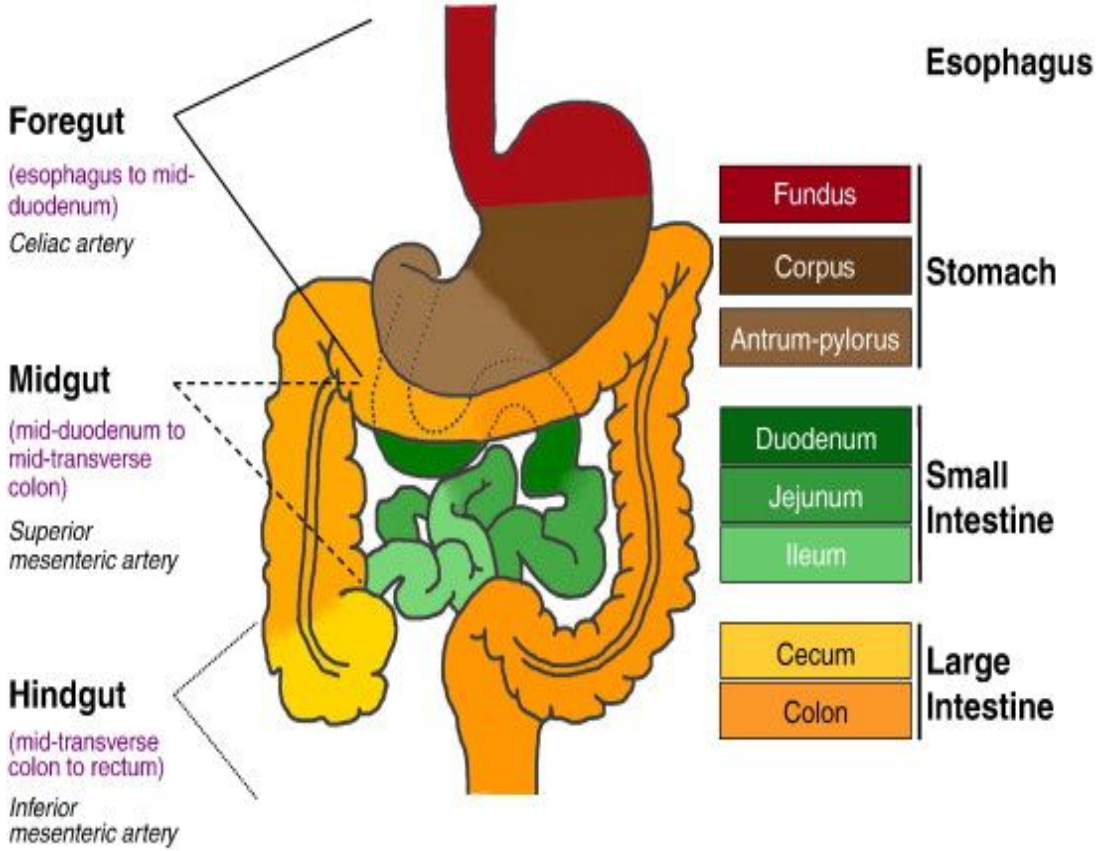


Arterial Supply of the Primitive Gut:

- The primitive gut tube can be divided into ***foregut, midgut, and hind gut.***
- The boundaries of foregut, midgut, and hindgut are defined by the areas supplied by the three branches of abdominal aorta.

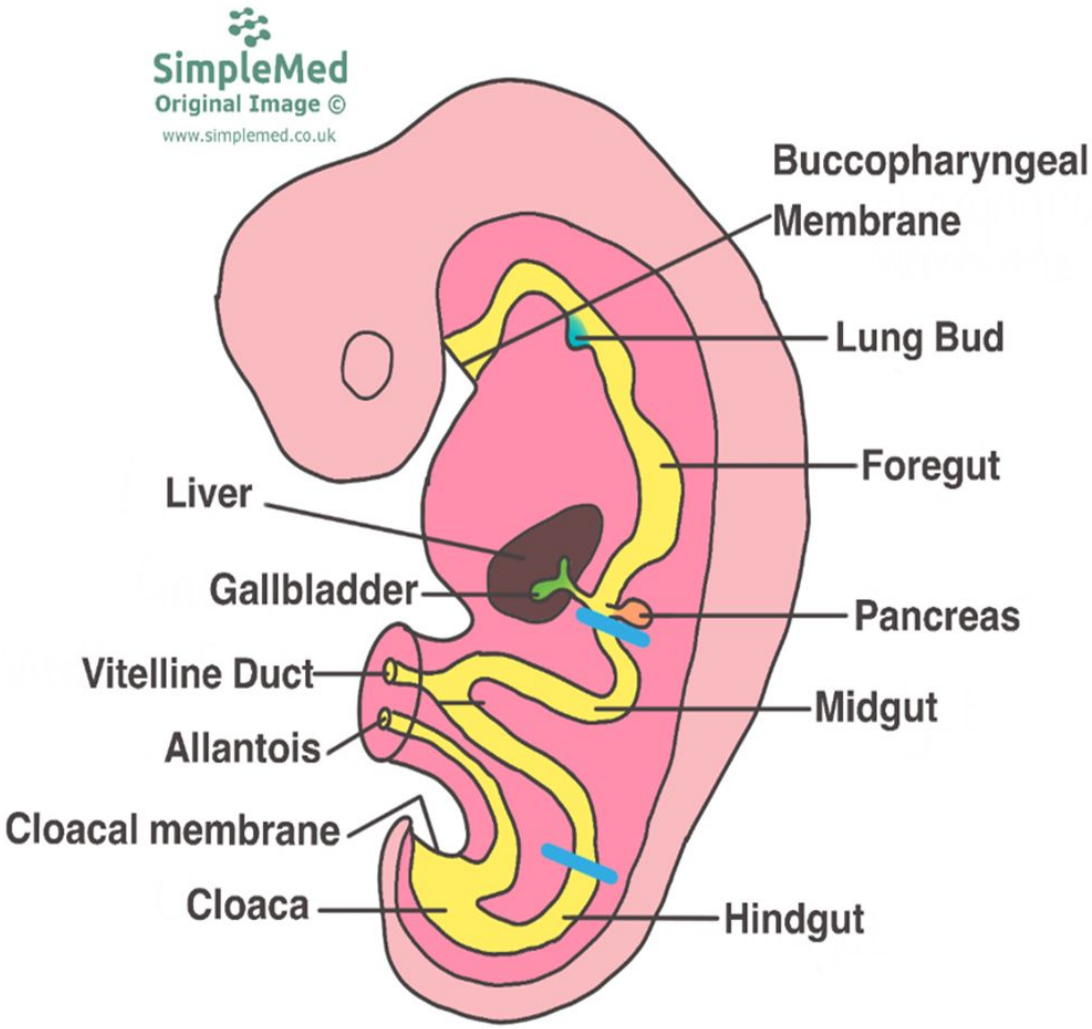
Foregut:

- It begins with the abdominal part of esophagus and ends just inferior to major duodenal papilla.
- It includes ***abdominal esophagus, stomach, duodenum (superior to major duodenal papilla), liver, pancreas, and gall bladder.***
- It is supplied by **coeliac trunk.**



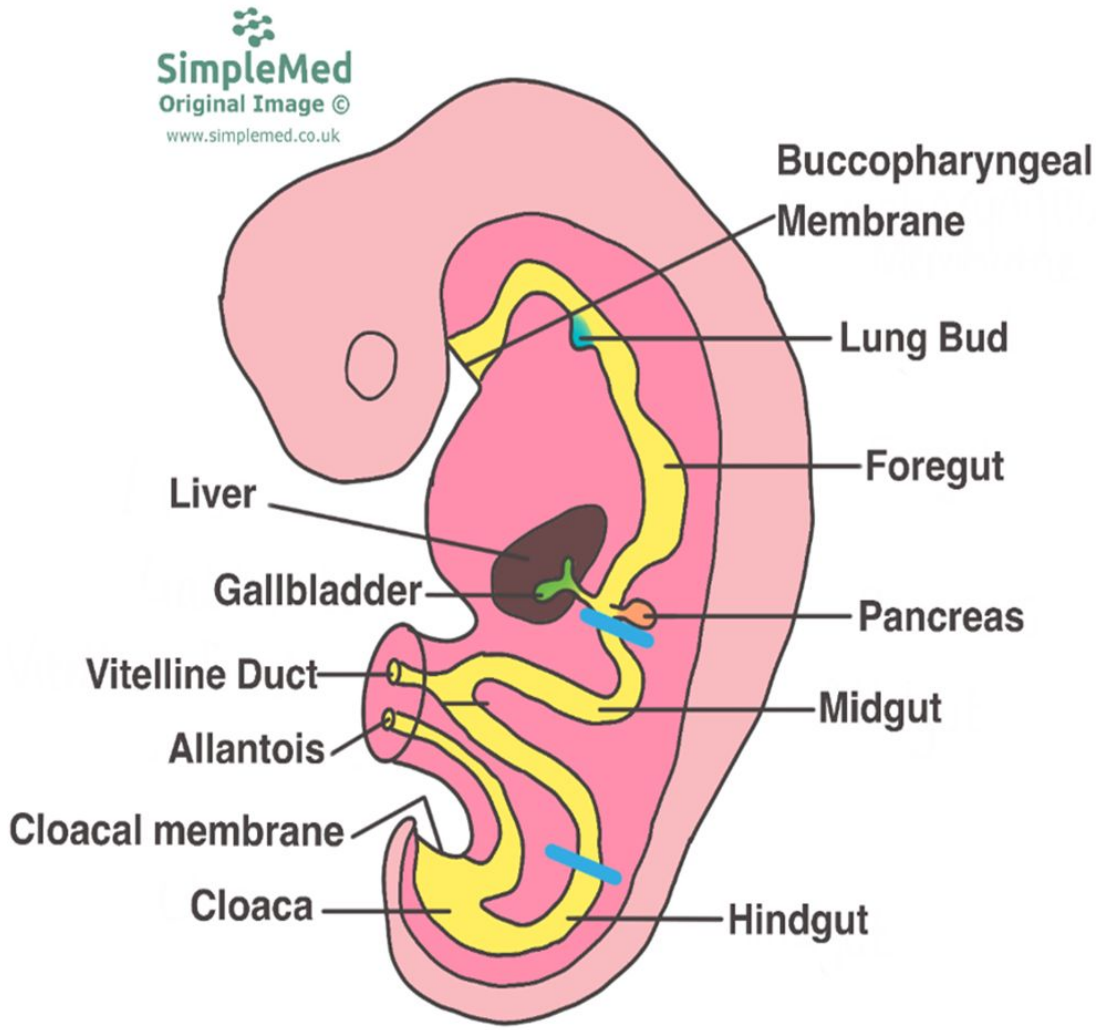
Midgut:

- It begins just inferior to major duodenal papilla and ends at the junction between proximal 2/3rd and distal 1/3rd of transverse colon.
- It consists of ***duodenum (inferior to major duodenal papilla), jejunum, ileum, cecum, appendix, ascending colon, and right 2/3rd of transverse colon.***
- It is supplied by **superior mesenteric artery.**

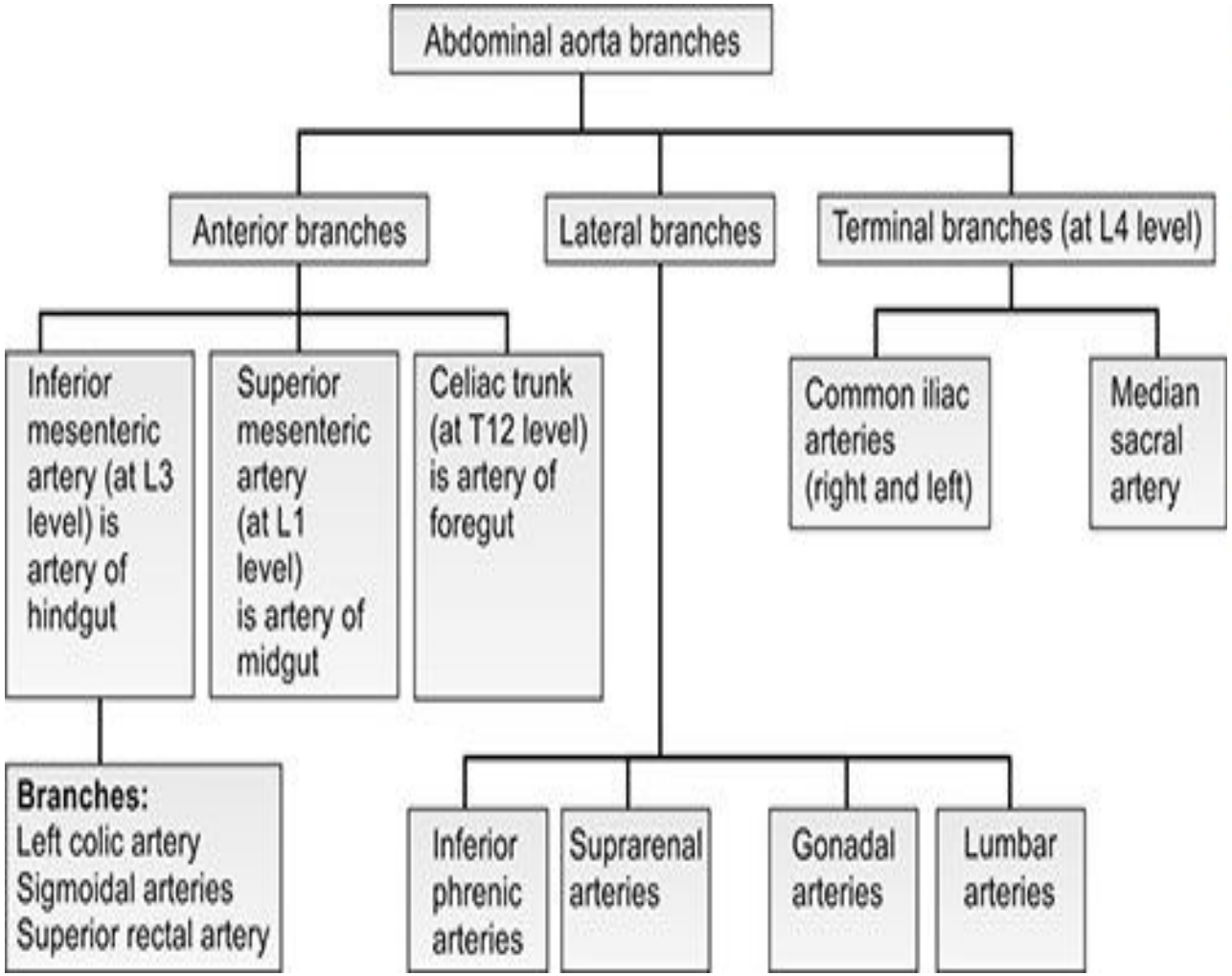


Hindgut:

- It begins at the junction between proximal 2/3rd and distal 1/3rd of transverse colon, and ends midway through the anal canal.
- It includes ***left 1/3rd of transverse colon, descending colon, sigmoid colon, rectum, and upper half of anal canal.***
- It is supplied by **inferior mesenteric artery.**



Abdominal Aorta & its Branches:



Note

Abdominal Aorta Branches

Visceral branches

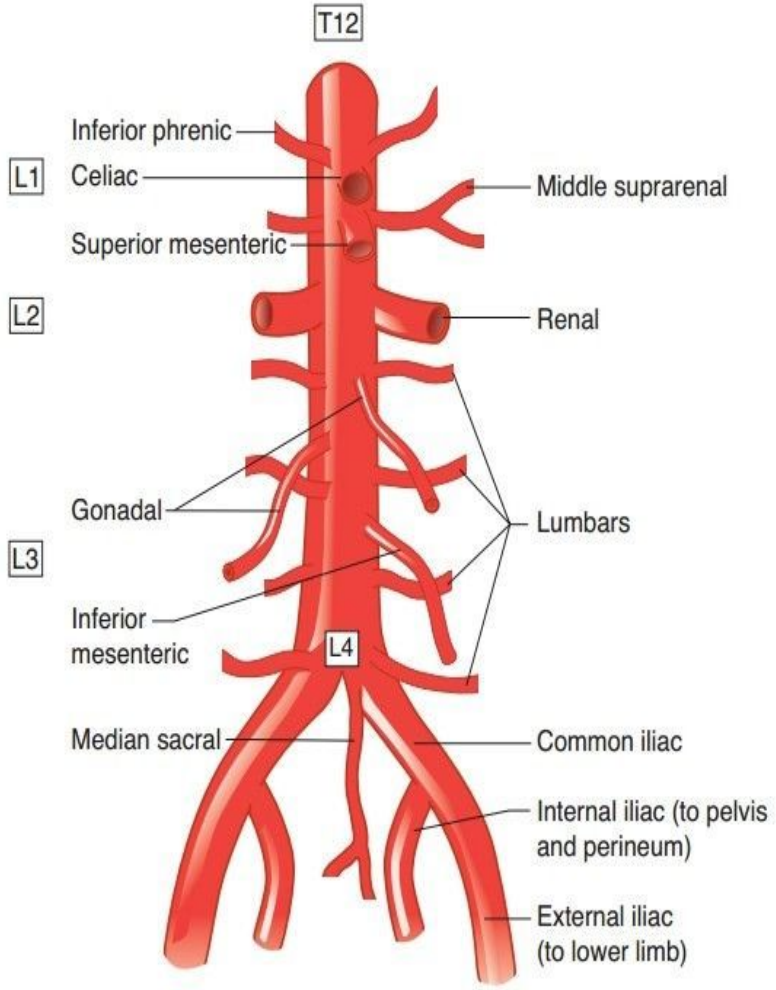
Unpaired: celiac (foregut), superior mesenteric (midgut), inferior mesenteric (hindgut)

Paired: middle suprarenals, renals, gonadals

Parietal branches

Unpaired: median sacral

Paired: inferior phrenics, lumbars, common iliac



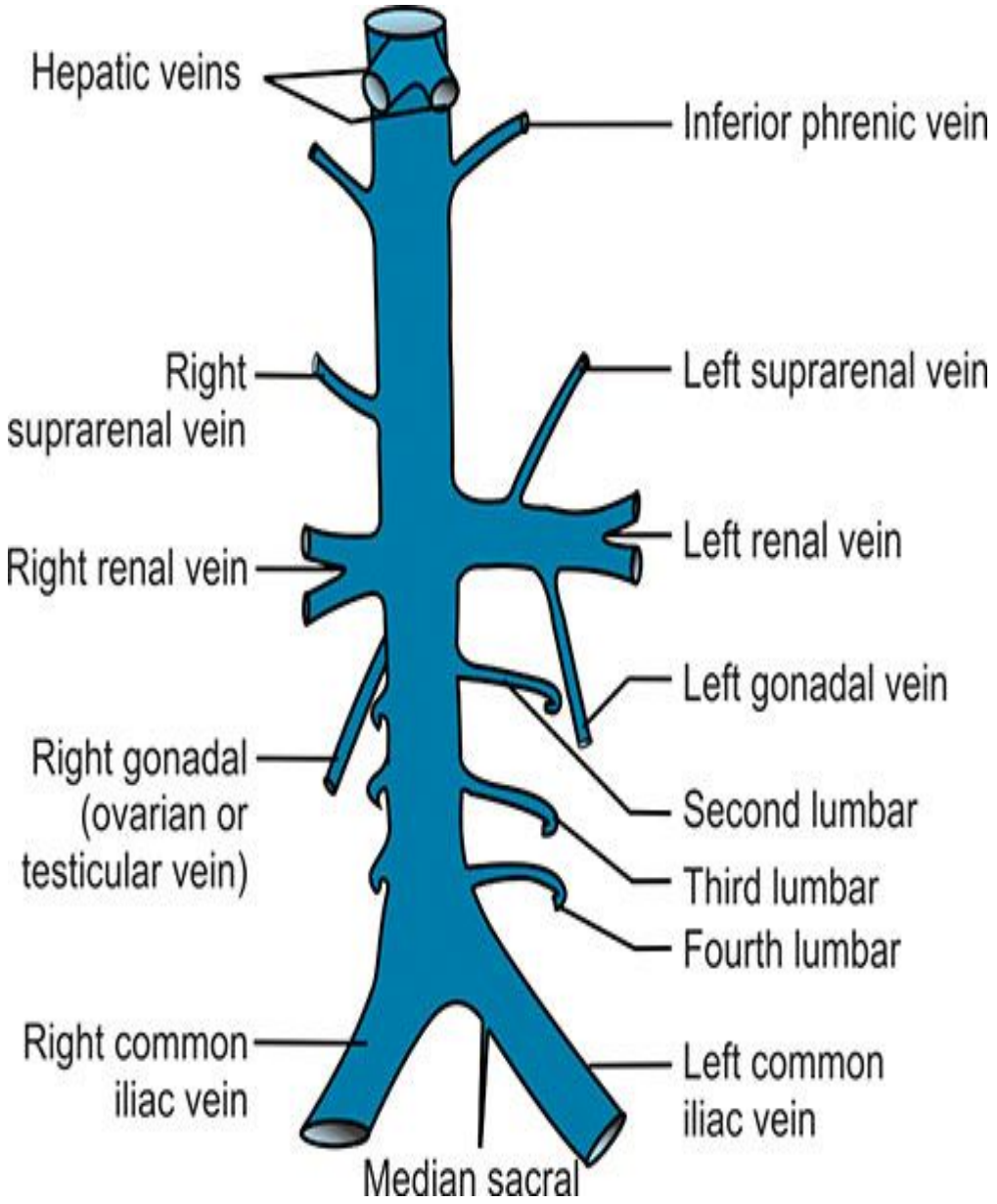
INFERIOR VENA CAVA

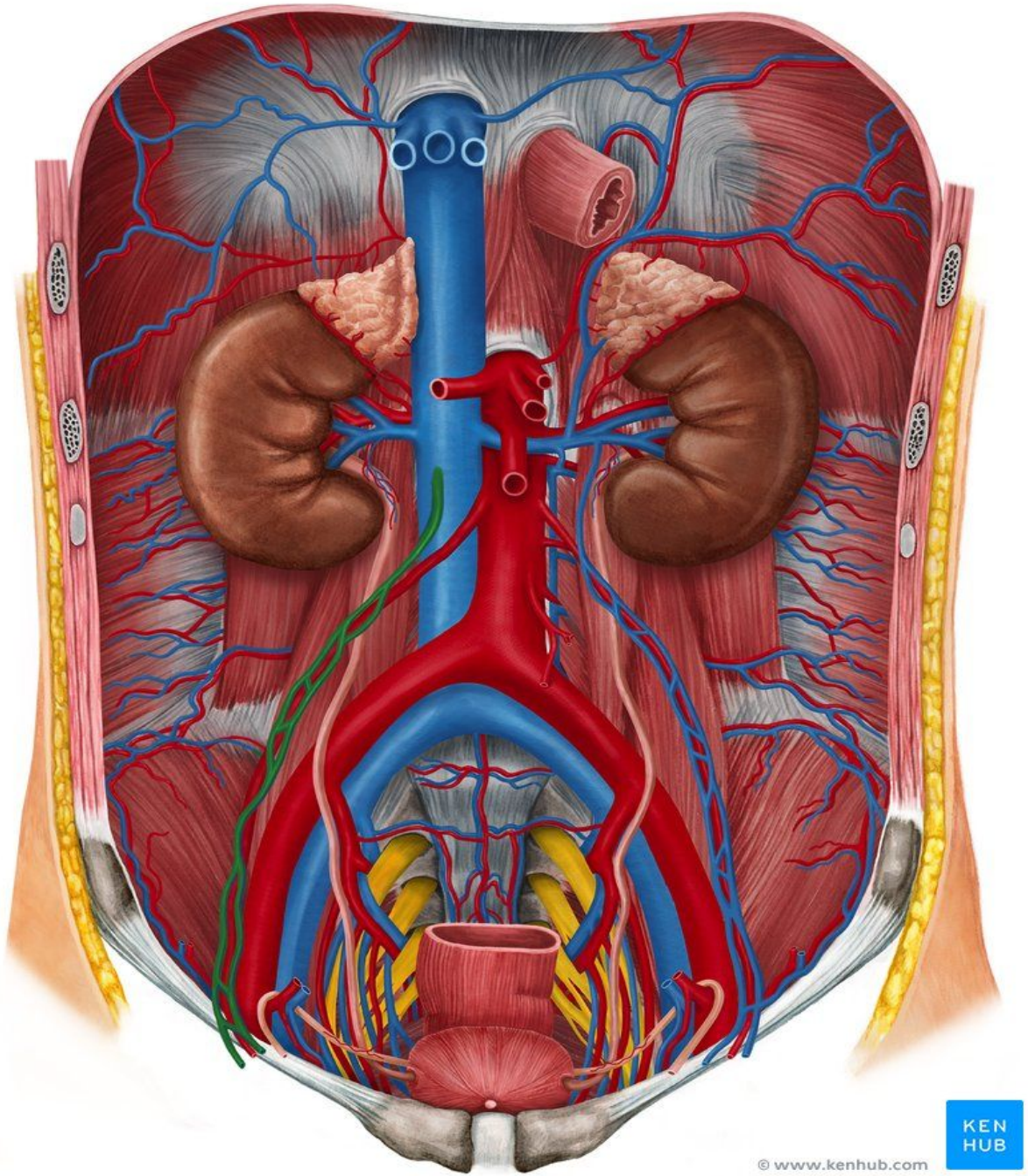
❑BEGINNING:

- Formed by union of 2 common iliac veins (L5)

❑TERMINATION:

- Pierces central tendon of diaphragm at level of T8
- Opens in the right atrium at level of right 6th costal cartilage





Tributaries of Inferior vena cava

mnemonic : I Like To Rise So High

Iliac vein (common)

Lumbar vein

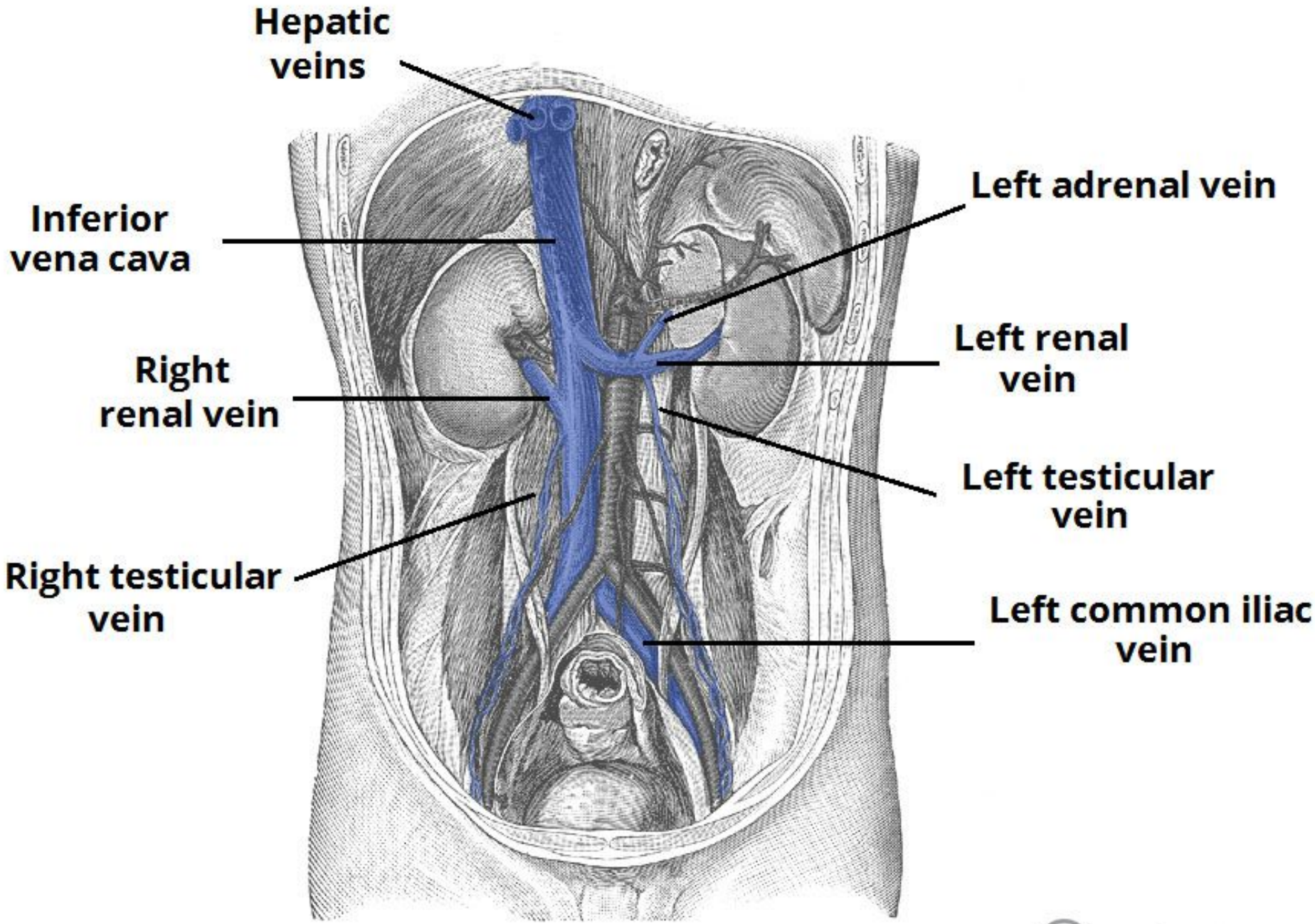
Testicular (gonadal) vein

Renal vein

Suprarenal vein

Hepatic vein

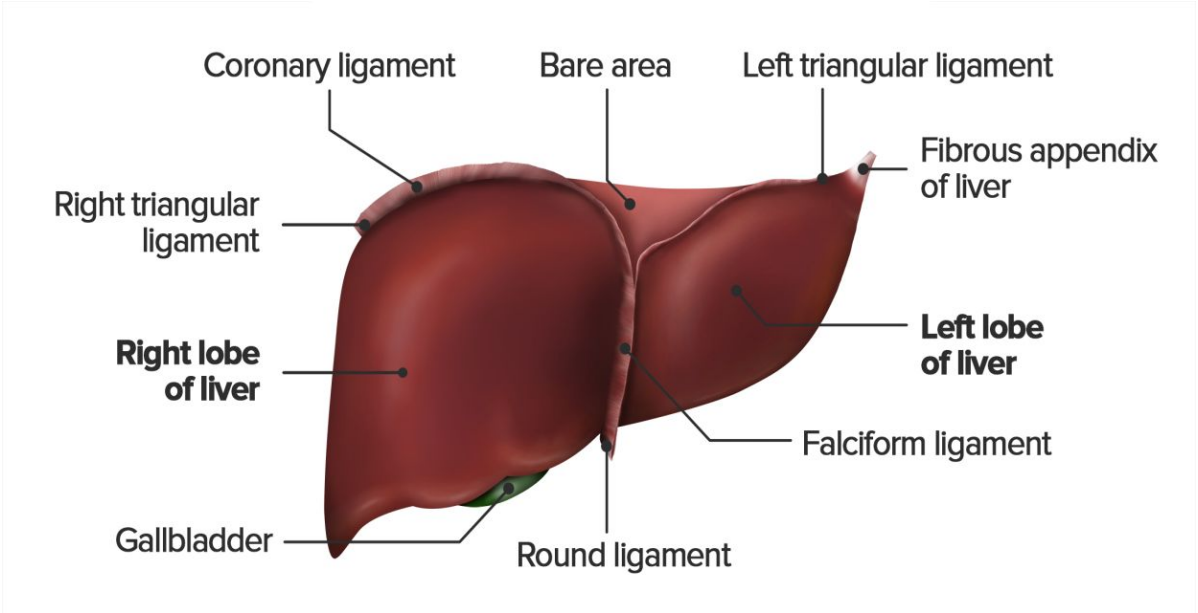


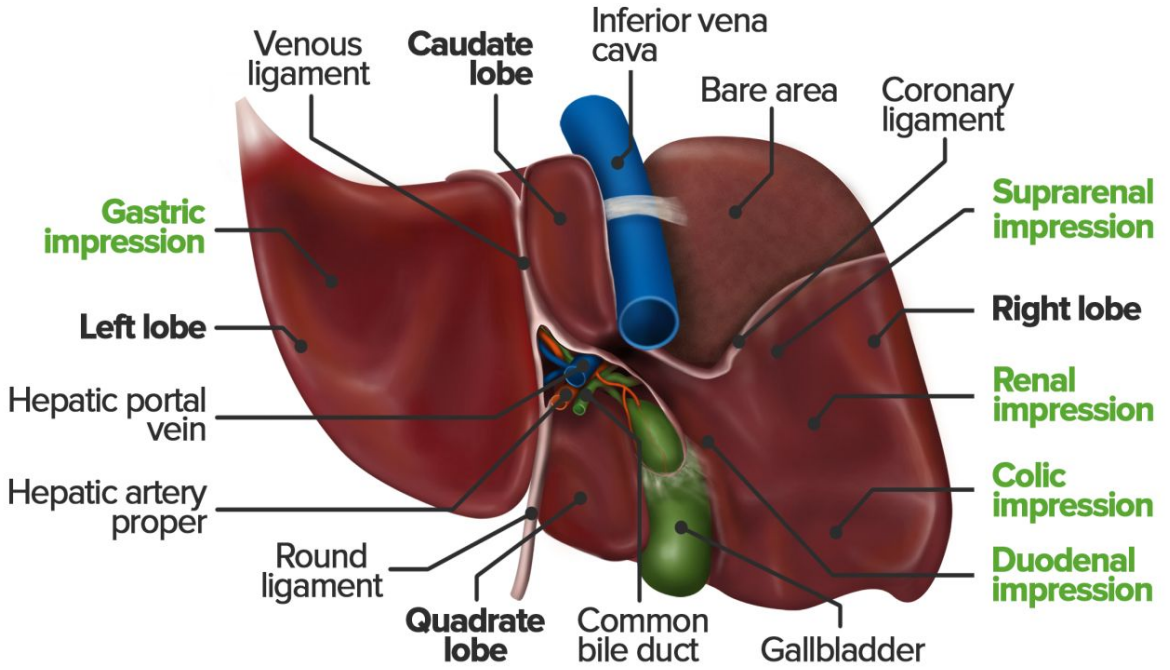


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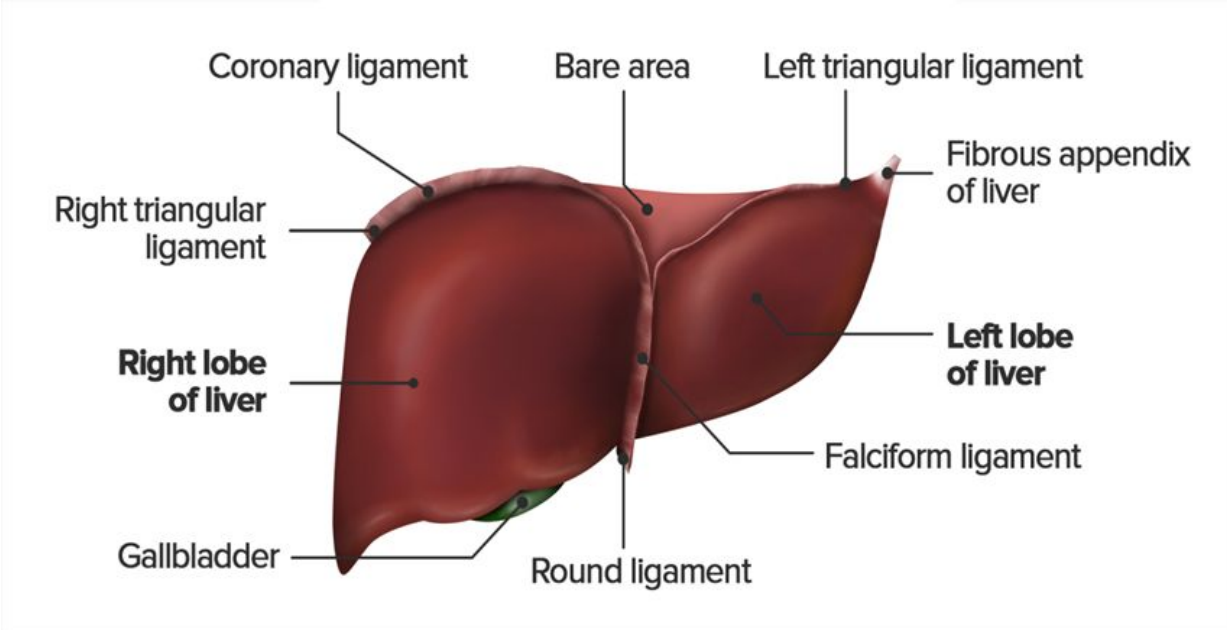
Gross Anatomy of Liver:

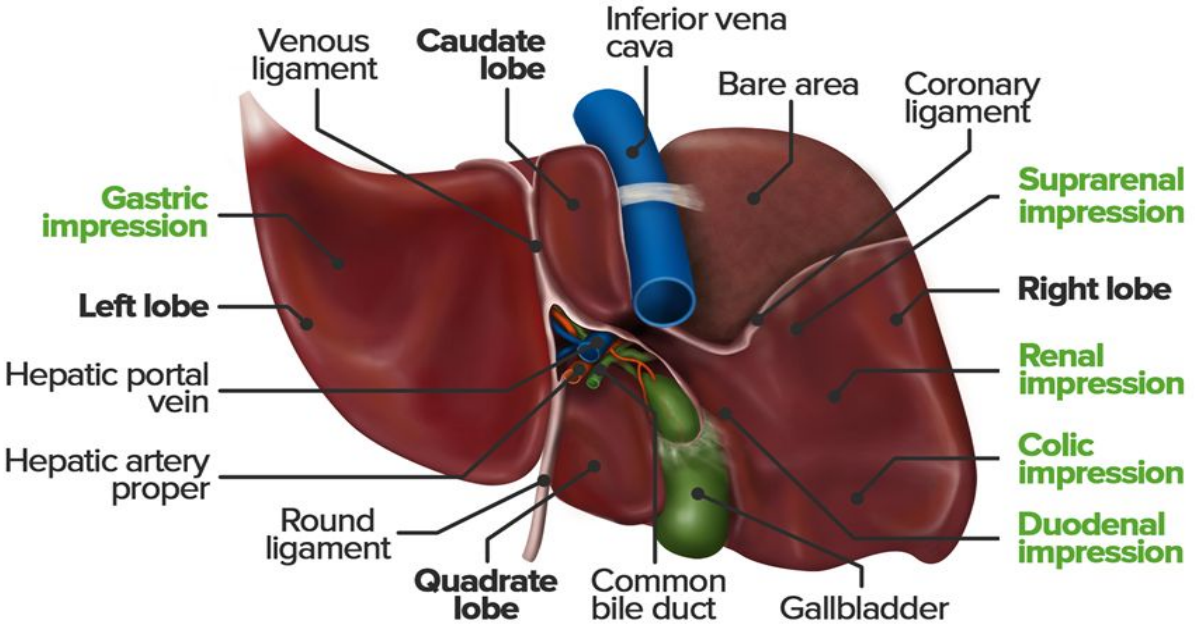
- Liver is the largest visceral organ in the body.
- It mainly occupies right hypochondrium, epigastrium, and a little extending to the left hypochondrium.
- Surfaces of the liver include:
- A **diaphragmatic surface:** it is smooth and lies against inferior surface of diaphragm, and
- A **visceral surface:** it is covered with visceral peritoneum except fossa for the gall bladder and porta hepatis. It is related to the following structures:
- **Esophagus**
- **Stomach**
- **Duodenum**
- **Gall bladder**
- **Right colic flexure & right transverse colon**
- **Right kidney & adrenal gland**



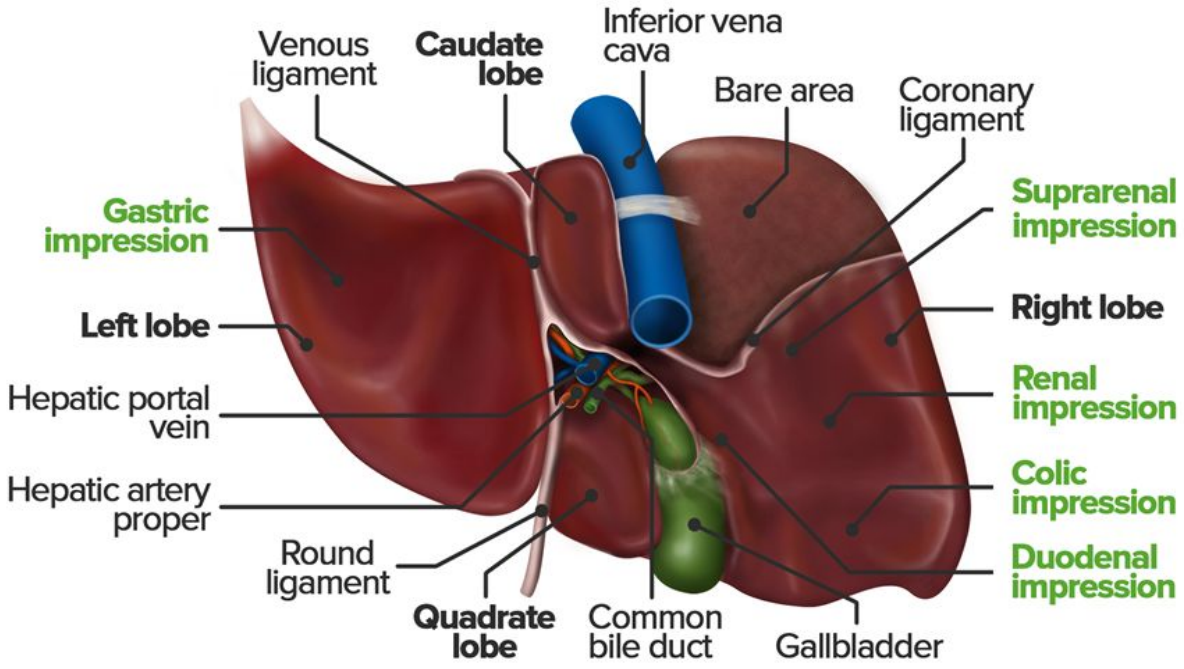


- The porta hepatis/gateway of liver contains hepatic arteries, portal vein, and bile duct.
- The liver is attached anteriorly to the abdominal wall by falciform ligament.
- The liver is completely covered by visceral peritoneum except for ‘bare area’.
- Bare area is a part of liver on diaphragmatic surface where there is no intervening peritoneum between liver and diaphragm.



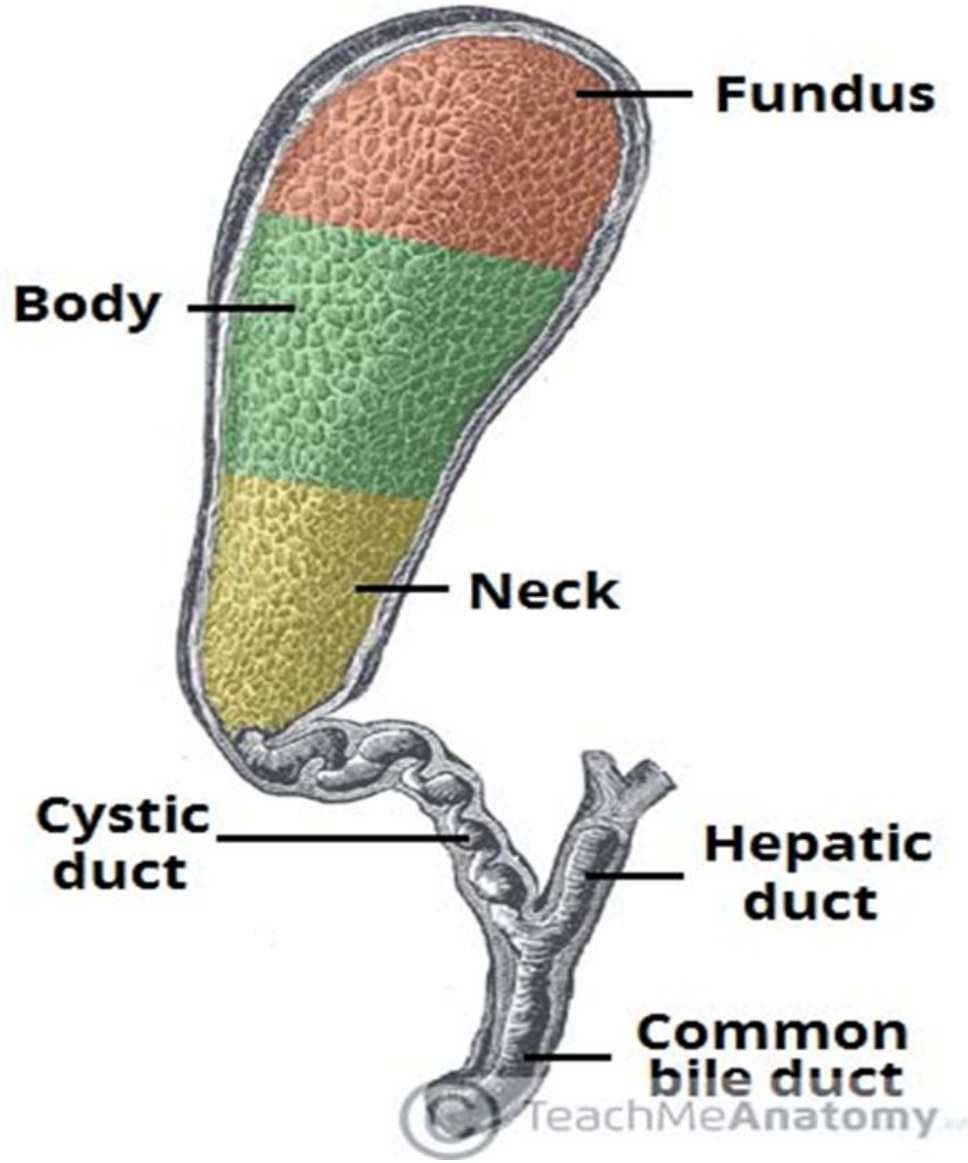


- The liver is divided into right and left lobes by fossae for gall bladder and inferior vena cava.
- The right lobe is the largest.
- The quadrate lobe is visible on the anterior aspect of visceral surface. It is bounded on the left by fissure for ligamentum teres (round ligament) and on the right by fossa for gall bladder.
- The caudate lobe of liver is visible on the posterior part of visceral surface. It is bounded on the left by fissure for ligamentum venosum and on the right by groove for inferior vena cava.

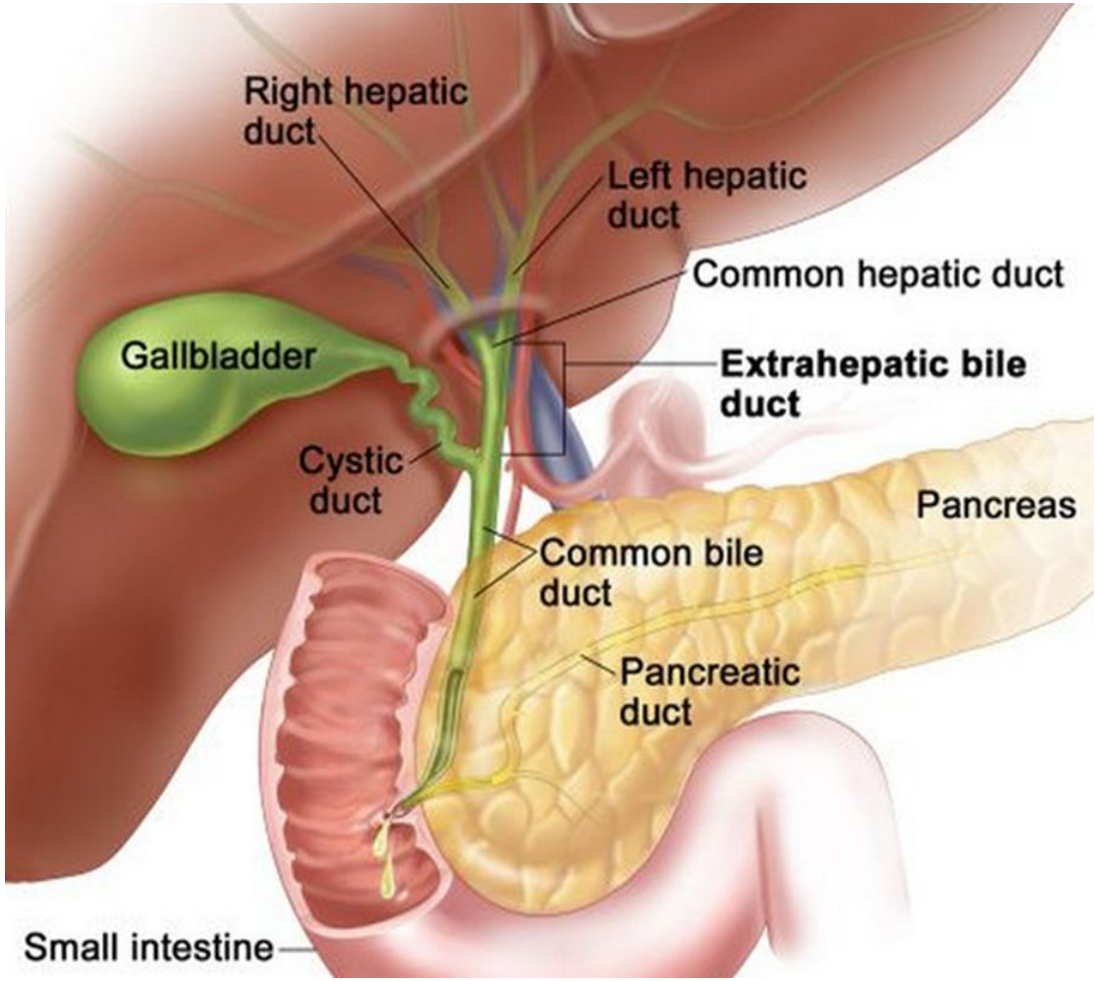


The Gall Bladder:

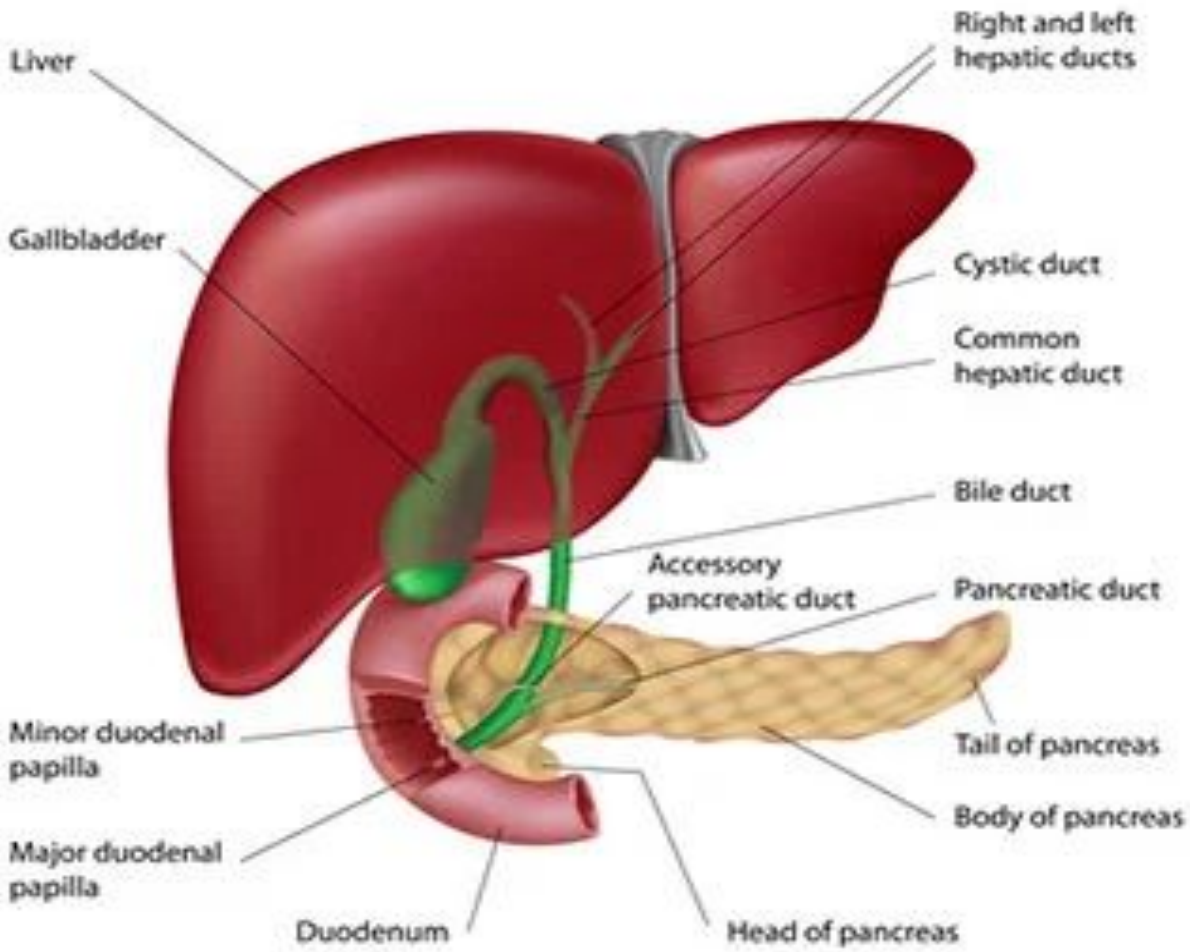
- The gall bladder is a distensible, muscular sac which is attached to the visceral surface of liver.
- Its main function of gall bladder are concentration and storage of bile.
- The gall bladder is divided into **fundus, body, & neck.**
- It is lined by simple columnar epithelium.
- The wall of gall bladder consists of 3 coats, the **mucosa, muscularis, and serosa/adventitia** (it does not have a submucosa).
- **Cystic duct:** connects the gall bladder with the with the common hepatic duct (from liver) to form ***common bile duct (CBD).***

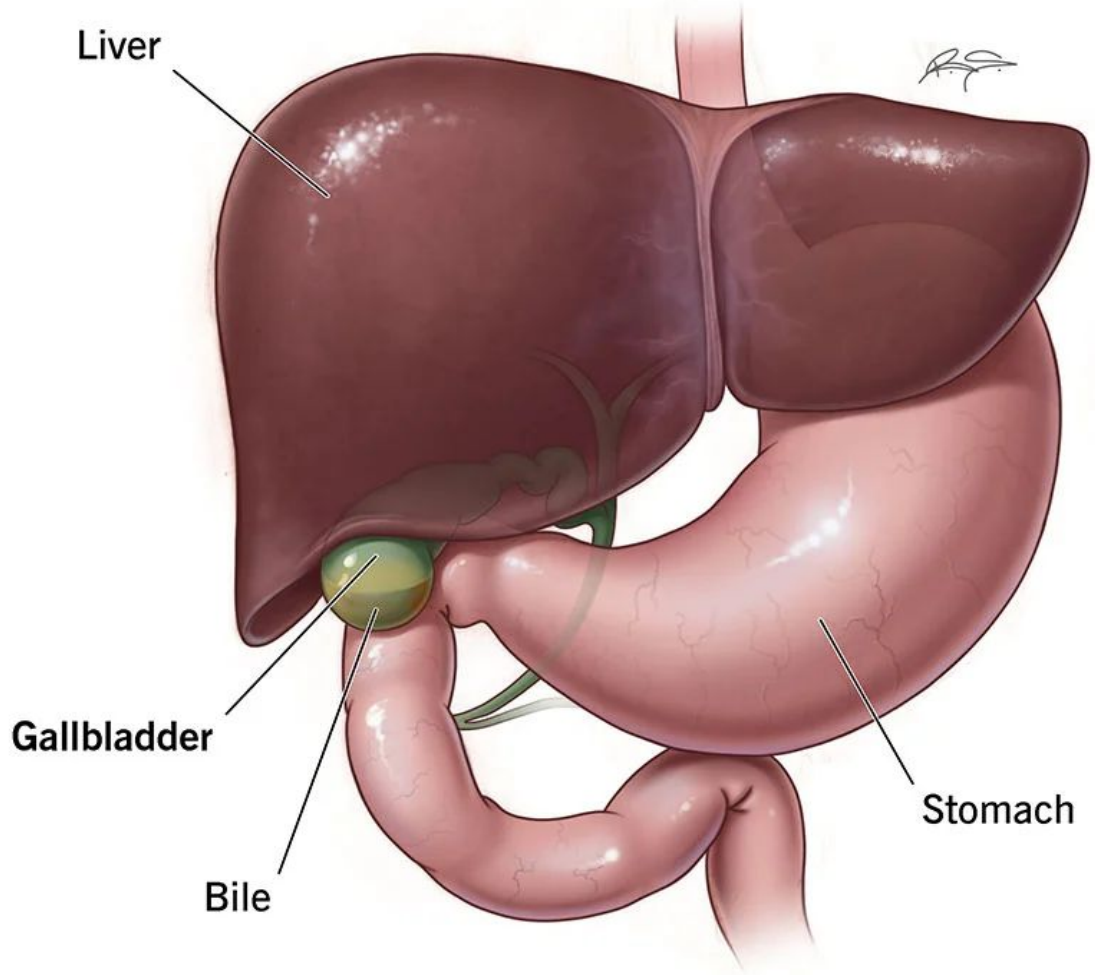


- The CBD along with the major pancreatic duct opens into the summit of major duodenal papilla in the 2nd part of duodenum.
- **Arterial supply:** cystic artery (a branch of right hepatic artery).



Liver, Gallbladder, Pancreas and Bile Passage





Hepatic Portal Vein

- Origin & end in capillaries / venous sinusoids
- Size: 8cm X 1 cm
- Drains
  - Abdominal part of alimentary tract (except lower part of anal canal)
  - Spleen & Pancreas
- Conveys absorbed products of digested food to liver
- Devoid of valves
- Reservoir of blood : 1200 ml / min



Hepatic Portal Circulation

- Portal Vein: carries blood from one capillary network to another

- Hepatic Portal Vein
  - Formed by union of splenic and superior mesenteric veins
  - Receives blood from digestive organ capillaries and delivers it to liver capillaries

